

Consequently, in a population lying initially above the line, both the trait and the preference for it would increase indefinitely, until some factor not allowed for in the model intervenes. Lande interprets this as illustrating Fisher's runaway process.

Further Reading

Bradbury, J.W. and Andersson, M. (ed.) (1987). *Sexual selection: testing the alternatives*. Wiley, New York.

Bull, J. (1983). *Evolution of sex determining mechanisms*. Benjamin Cummings, Menlo Park, California.

Charnov, E.L. (1982). *The theory of sex allocation*. Princeton University Press.

Problems

Part A In answering these questions on the sex ratio, assume (what may not be the case) that there are genes which, acting in a female, can alter, without cost, the sex ratio of her offspring at conception. For each of the following cases, do you expect a bias in the sex ratio at conception: if so, in favour of which sex? (N.B. the facts stated are only roughly true.)

1. In red deer, a hind is more likely not to breed if she produced a son in the previous year than if she produced a daughter.
2. In seals, the litter size is one. Males and females are the same weight at weaning, but males are more likely to die between weaning and sexual maturity.
3. In a mammal species with a litter size of 5-8, there are many sex-linked genes that cause death before implantation.
4. As Question 3, but the only effect of sex-linked recessives is to cause male sterility.
5. In a butterfly species (females heterogametic), there are many sex-linked recessives causing eggs to die before hatching.
6. A social spider lives in endogamous colonies, that produce new colonies by fission: there is no migration between colonies.
- 7.* An imaginary carnivore (not unlike a lion) produces litters of 2. Brothers cooperate in finding mates, so a male has a higher reproductive success if he has a brother. If a female has complete control over her sex ratio, what would you expect to happen?

Part B The answers to these questions are highly speculative: in each case, suggest an explanation for the observation, and, if possible, suggest further observations that might confirm or refute it.

1. Most annual weeds are self-fertile hermaphrodites.
2. Some plant species are 'gynodioecious': that is, there are hermaphrodites and females. Femaleness (that is, male sterility) is often caused by a cytoplasmic gene, transmitted in the ovule but not in the pollen.
3. Regular inbreeding is commoner in haplo-diploid species than in ecologically similar diploid ones.

4. Dioecy in plants is usually interpreted as an adaptation to prevent selfing; it is commoner in trees than in herbs.
5. Allopolyploidy (see Chapter 14) as a mechanism of speciation is rare in animals.

Computer Projects

1. Sexual selection. Consider a haploid sexual species with two alleles at each of two loci T and C . T males have a long tail, and t males a short tail: their chances of survival are in the ratio 1 s:1. C females prefer to mate with T males, whereas c females mate randomly. The allele T has no effect in females, and allele C has no effect in males. First, devise a plausible formula for the frequencies of different kinds of mating, given the frequencies of T and C , and the fact that C females prefer T males. (Several different models are possible, giving different formulae: you may like to try more than one, but stick to simple formulae.) How do 'choice' and 'tail length' evolve? How does the final state of the population depend on initial conditions?
2. In a self-sterile hermaphrodite plant, a mutant A arises such that Aa and AA are self-fertile: they pollinate all their own seeds, and contribute as much pollen to the pollen pool as self-sterile aa plants. The viability of offspring produced by selfing is V_1 if the parent came from an outcross, and V_2 if the parent was itself the result of selfing. Ignore differences in fecundity. Model the evolution of such a population. If $V_1 = V_2 = V$, what value of V permits A to spread when rare? What value of V permits A to go to fixation? If $V_2 < V_1$, is a stable polymorphism possible?

**Chapter 14—
Macroevolution.**

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This book has been concerned with processes that can be studied in contemporary populations over relatively short time-periods. Our picture of evolution on a larger scale (macroevolution) comes from comparative anatomy and embryology, from taxonomy and geographical distribution, and from palaeontology. The question naturally arises whether the processes of population genetics are sufficient to account for macroevolution. Very different views can be held on this question (see, for example, the books by Simpson and by Stanley in the Further reading section at the end of this chapter). I now briefly review some macroevolutionary phenomena with this question in mind.

Species and Speciation

Animals and plants are classified into species. Do the species of our classifications correspond to real divisions in the world, or are they merely arbitrary groupings that we need in order to communicate with one another? Are there really blue, great, coal, marsh, and willow tits in British woodlands, and western gray, dusky, and Hammond's flycatchers in the western US? In most cases (including the examples just given), the groups are real. Perhaps the most convincing evidence is the fact that pre-literate people often recognize the same species (and sometimes the same higher categories) as do modern taxonomists. If species were arbitrary divisions, this would not be so.

Three reasons have been suggested for the existence of species:

1. Only certain kinds of organism are possible. Thus consider the analogy with chemistry. There exist only a limited number of distinct elements, and a larger number of distinct compounds, because only those configurations of matter are stable. It has sometimes been argued that there are similar, but unknown, laws of biological form that permit only a limited number of distinct species. I mention this view because it has often been proposed, but I know of no

sensible reason for holding it: the facts of geographical distribution alone seem sufficient to disprove it.

2. The number of distinct ecological niches is finite, and determines the kinds of species that can exist. In a few cases, this may be true. For example, the wydah birds of Africa (*Vidua*) are brood parasites of Estrildid finches, and are usually species-specific. The nestlings mimic the pattern of spots in the gape of the host species, revealed when the nestling begs for food. The pin-tailed wydah parasitizes several species of waxbills, whose mouth patterns are the same. Thus the number of wydah species is limited by the number of potential host species. But this cannot be convincing as a general explanation. One reason is the absence of clear species distinctions in some parthenogenetic complexes, such as *Abies* (hawkweeds) and *Rubus* (brambles).

3. Species exist because organisms reproduce sexually. Blue tits are genetically similar to one another (and different from great tits) because they are descended from a common set of interbreeding ancestors (and great tits are descended from a different set). The two species will remain distinct in the future because they do not interbreed.

There is little doubt that the third explanation is correct. It follows that if two contemporary populations have no opportunity to interbreed, and have had no such opportunity for many generations, then it is largely a matter of convention whether we place them in the same species. The difference between such geographically isolated species may be as great as that between distinct sympatric species (**sympatric** means living in the same place; **allopatric** means living in different places), or there may be no distinguishable difference, or the degree of difference may lie anywhere between these extremes. This is what we would expect from explanation (3) above, but not from explanations (1) or (2).

This point is nicely illustrated by the birds of North America. In a number of cases (e.g. flickers, meadow larks, towhees, bluebirds, kingbirds, peewees) there are similar but distinguishable populations on the east and west coasts. Are these distinct species, or merely varieties? There are three possibilities:

1. There is a continuous series of interbreeding populations connecting the two forms. The populations are then treated as belonging to a single species.
2. There is a region of overlap between the eastern and western populations, in which they do not interbreed, but remain distinct. The populations are then placed in different species.
3. The two populations are geographically isolated. The decision as to whether they should be placed in one species or two is then a matter of convention.

It is widely agreed that the differences between species usually originate during geographical isolation: that is, speciation is allopatric rather than sympatric. The reason for this is that species typically differ at many gene loci: in the absence of geographic isolation, such a polygenic difference would be broken down by

hybridization more rapidly than it could be built up. It is in fact possible to devise processes that could lead to sympatric speciation (e.g. Maynard Smith 1966). In effect, they involve the establishment of a strongly selected single-gene polymorphism (for example, by Levene's mechanism see p. 71), followed by the association of mating isolation with that polymorphism. Such processes may occasionally occur, but no doubt allopatric speciation is the rule.

Table 14.1 classifies the processes, known as **isolating mechanisms**, that prevent the flow of genes between sympatric species. There is an important distinction between pre- and post-zygotic mechanisms. An individual that mates with a member of another species, and then produces inviable or infertile offspring, has wasted its reproductive effort. A gene causing it to be more likely to mate with a member of its own species would therefore be favoured by selection. This has led to the suggestion that pre-zygotic isolating mechanisms are the result of natural selection favouring isolation. I now discuss this possibility.

It is common to find that the most striking differences between closely related sympatric species concern characters used in mate recognition. For example, in Britain the three leaf warblers 柳莺 warbler, chiffchaff, and wood warbler 槲 are hard to tell apart in the field except by their songs. Closely related *Drosophila* species are often only distinguishable morphologically by examining their genitalia. However, such observations can be explained in two ways:

1. The **reinforcement** hypothesis. Genetic divergence in allopatry has reduced the viability or fertility of hybrids. After subsequent sympatry, initially slight differences in mate recognition traits have been exaggerated by selection in favour of pre-zygotic isolation.

Table 4.1

Factors that prevent gene flow between species

- I. Geographical
Species live in different places
- II. Factors preventing the formation of hybrids (pre-zygotic factors)
 - Plants*
 - A. Isolation by habitat: ecological isolation
 - B. Different flowering seasons
 - C. Pollination constancy by insects
 - D. Lack of pollen-tube growth down style of a different species
 - Animals*
Differences in courtship behaviour, and in signals used in mate recognition.
- III. Factors acting after the formation of hybrids (post-zygotic factors)
 - A. F_1 hybrids inviable
 - B. F_1 hybrids infertile
 - C. Hybrid breakdown: that is, F_1 hybrids viable and fertile, but F_2 and backcross hybrids inviable or infertile

2. Traits concerned in mate recognition evolve particularly rapidly, even in allopatry: pre-zygotic isolation evolves in allopatry, and not in response to subsequent sympatry and hybridization. There is, I think, no accepted term for this hypothesis: I shall call it the 'recognition in allopatry' hypothesis.

The best data bearing on the problem are those analysed by Coyne and Orr (1989) on 118 pairs of closely related species of *Drosophila*. For each pair, the following were estimated:

1. **Genetic distance.** This is an estimate, devised by Nei (1972), of the number of amino-acid substitutions in proteins that separate two species, based on electrophoretic data. Since rates of molecular evolution are, very approximately, constant, the genetic distance is an estimate of the time since divergence. One reason for thinking that it is a reliable estimate is as follows. Phylogenies (family trees) of the genus *Drosophila* based on genetic distance are almost identical to those based on inversions (see p. 223): since inversions are unique and irreversible events, they are the most reliable basis we have for the construction of phylogenies.

2. Post-zygotic isolation: that is, degree of hybrid inviability and infertility. Hybrid inviability was measured as follows. If both sexes, from both reciprocal crosses, were completely inviable, the measure = 1; if both sexes, from both reciprocal crosses, were at least partially viable, the measure = 0; intermediate values of 0.25, 0.5, and 0.75 imply that 1, 2, or 3 of the four possible types (two sexes $\times$ two crosses) were inviable. A similar measure of sterility was used. It was found that hybrid inviability and infertility evolve at much the same rate.

3. Pre-zygotic isolation: that is, the degree of mating isolation, measured as $(1 - \text{frequency of matings between species} / \text{frequency of matings within species})$ in laboratory tests.

Mating discrimination, sterility, and inviability all increase gradually with time, as estimated by genetic distance. This confirms the impression gained from a study of geographical variation that speciation (at least, in *Drosophila*) is not a sudden event. Among allopatric species, pre- and post-zygotic isolation evolve at similar rates. For sympatric species, however, pre-zygotic isolation evolves more rapidly (Table 14.2). For species pairs with low genetic distance, strong pre-zygotic isolation is found *only* in sympatric pairs.

Table 14.2

Degree of isolation between pairs of *Drosophila* species
(Coyne and Orr 1989)

	Allopatric	Sympatric
Pre-zygotic isolation	0.25	0.62
Post-zygotic isolation	0.32	0.34

How are these results to be interpreted? We should not conclude, from the fact that mating discrimination evolves first in sympatric pairs, that the whole speciation process is sympatric. It is much more likely that the first steps took place in allopatry. However, we still have to choose between the two hypotheses listed above: reinforcement, or recognition in allopatry. I will consider them in turn:

1. The reinforcement hypothesis. This does predict the data of Table 14.2. There is, however, one fact that is awkward, at least at first sight. This is that some sympatric pairs show pre- but not post-zygotic isolation. Clearly, if there is no post-zygotic isolation, there is no selection favouring mate discrimination, so reinforcement cannot happen. But the difficulty may be more apparent than real. The measure of post-zygotic isolation used requires the complete inviability or sterility of at least one of the hybrid types: this is a very extreme requirement. There must be many cases in which the hybrids show some degree of lowered fitness in the wild that does not show up in the data as analysed. It will be important to look again at those cases that show pre- but not post-zygotic isolation.
2. Recognition in allopatry: that is, mate discrimination evolves more rapidly than post-zygotic isolation, even in allopatry. This hypothesis does predict some difference between sympatric and allopatric pairs, for the following reason. A pair of populations that have not evolved a fair degree of isolation, either pre- or post-zygotic, while allopatric are likely to fuse into a single interbreeding population if their ranges later come to overlap. Therefore we only expect to observe sympatric pairs that do show a fair degree of isolation, whereas allopatric pairs could show little or no isolation. But this does not really explain the data. If the 'recognition in allopatry' hypothesis were correct, we would expect to find that allopatric pairs, as well as sympatric ones, showed a greater degree of pre- than of post-zygotic isolation: in particular, we would expect to find some closely related allopatric pairs that show strong mate discrimination, and we do not.

The *Drosophila* data, then, support the reinforcement hypothesis. But the question is still open.

Sometimes, the hybrids between two species are sufficiently fertile to permit genetic analysis. We can then ask how many loci are responsible for particular differences. The number can be estimated by comparing the variability of the F_1 and F_2 hybrids. Thus, suppose that a difference is caused by alleles at a single locus: that is, one species is aa and the other AA . The F_1 is Aa , and is no more variable than the parental populations. The F_2 consists of $1aa:2Aa:1AA$, and is genetically more variable. Now suppose that a difference is caused by a very large number of independently assorting loci. Again, the F_1 is no more variable than either parental population. But now the F_2 is no more variable than the F_1 . Thus, at any single locus there are three possible genotypes. But the proportion of loci which are,

respectively, homozygous for the 'low' allele, heterozygous, and homozygous for the 'high' allele, will be the same in all individuals. For the same reason, if a number of people each toss a coin 10 000 times, all will get approximately the same proportion of heads.

Lande (1981*b*) describes how F_1 , F_2 and backcross data can be used to estimate the number of loci responsible for a phenotypic difference between two populations, and explains the assumptions on which the estimates are based. The estimates are minimum ones: one important reason for this is that separate loci may be linked, and therefore behave as if only one locus was involved. He applies the method to two species of Hawaiian *Drosophila* that differ in head shape by about eight phenotypic standard deviations in females, and to a still greater degree in males (Fig. 14.1). *D. silvestris* has a conservative head shape, and *D. heteroneura* has an extremely widened head. These species have a lek system of mating, and the wide head of male *D. heteroneura* probably evolved through female choice. The difference between females presumably evolved because most genes that affect males also affect females. Analysis of the F_1 and F_2 data on females suggest a minimum of six to eight separate loci: since there is no crossing over in males, and there are only five pairs of large chromosomes and one dot chromosome, the true number of separate loci may be substantially greater.

In other cases, a significant adaptive difference between two populations may be caused by one or by a few loci. Gottlieb (1984) suggests that this may be commoner in plants than in animals, and that differences between closely related species in structure, shape, orientation, and presence versus absence are frequently discrete, and appear to be governed by one or two genes. He quotes the columbine,

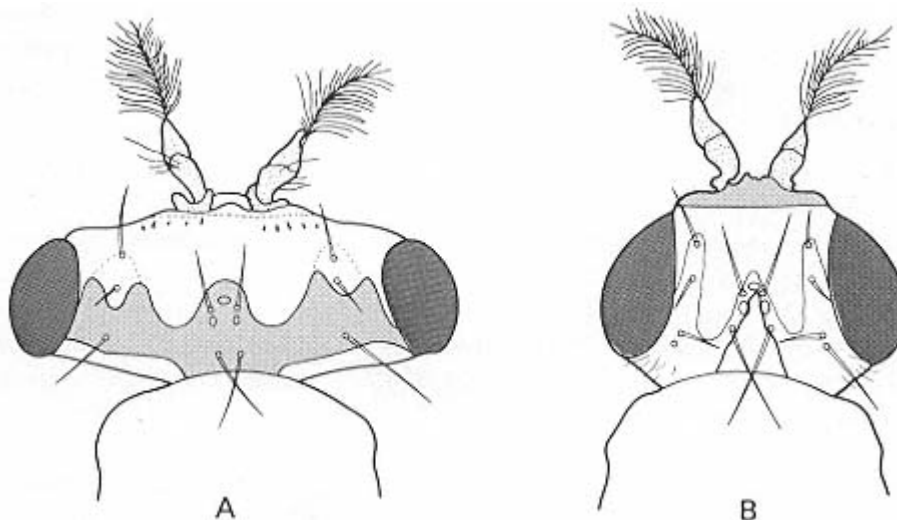


Figure 14.1

Head shape in **A** *Drosophila heteroneura* male, and **B** *D. silvestris* male. (From Val 1977.)

Aquilegia, as an example (Fig. 14.2). Members of this genus typically have nectar-containing spurs on each of the five petals, but in the primitive species *A. ecalcarata* these are replaced by small nectar-containing pockets. The F_2 between *A. ecalcarata* and other species indicates that the presence of a spur is caused by a single dominant gene. A second locus determines whether the spur is straight or curved at the tip (as in the figure), and a third locus whether the flowers are nodding (as in the figure) or erect. However, both length, and degree of curvature (if present), vary continuously, and are presumably polygenic. All these differences will affect which pollinators visit the flowers, and may therefore act as pre-zygotic isolating mechanisms.

What of the genetics of hybrid inviability and infertility? If a hybrid is inviable,

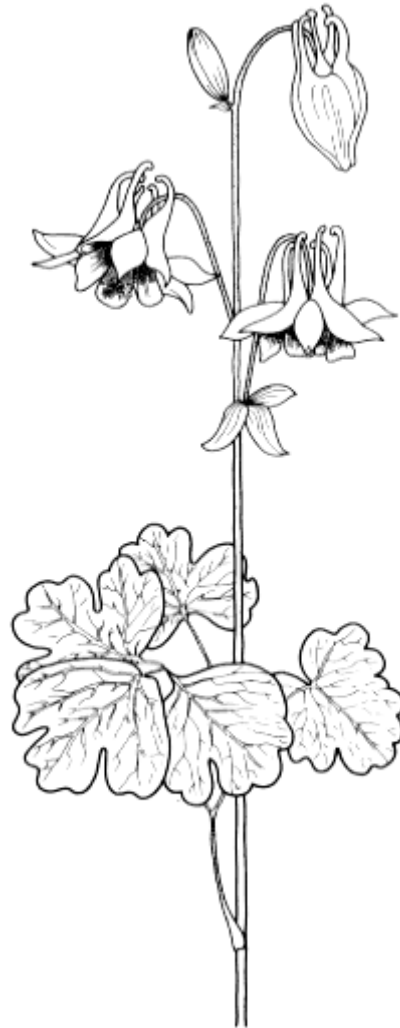


Figure 14.2
Aquilegia vulgaris.

does it follow that evolution has had to cross an adaptive valley? If so, it would imply that the process of speciation is in some way different from the typical hill-climbing process observed within populations. At the very least, it would suggest that the critical events took place by chance in a small population, as imagined in Wright's shifting balance model (p. 179). However, it was shown in Chapter 9 (Fig. 9.9, p. 179) that a large random-mating population can evolve by selection from state X to state Y , and yet the hybrids between X and Y can be of reduced fitness. In other words, hybrid inviability does not imply that an adaptive valley has been crossed. There is no reason to postulate atypical genetic events.

In practice, the events leading to a new species may often occur in small peripheral populations, because such populations are more likely to be exposed to an atypical environment and hence to directional selection, and to be sufficiently isolated spatially that genetic changes can occur without being swamped by gene flow from outside. There is little reason to think that small population size is itself important. However, some population geneticists would dissent strongly from this opinion: for an alternative view, see Templeton (1980).

So far, I have considered only cases in which both parental and daughter species are sexual diploids. In contrast, hybridization between two sexual species can give rise in one step to a new sexual tetraploid species, reproductively isolated from either parent. The process is known as allopolyploidy. Often, the diploid hybrid between two species is sterile, because the chromosomes from the two parent species fail to pair in meiosis. If, by chance, the chromosome number is doubled to give a tetraploid, each chromosome has one potential partner, and meiosis gives rise to regular diploid gametes, that can fuse to give a new tetraploid. The process is common in plants, both in domestication and in the wild, but rare in animals.

Patterns of Evolution

This section reviews some of the patterns detected by a study of the fossil record, and discusses how far they can be explained by natural selection within populations, and how far other mechanisms are called for.

The Rate of Evolution

First, we need a measure of the rate of evolution that is independent of scale: it should give the same estimate for an elephant that doubles in size in one million years (Myr) as for a fruitfly, and the same estimate for a species that doubles in size in one million years as for a species that quadruples (that is, doubles twice) in two million years. Haldane suggested that, if a measurement changes from x_1 to x_2 in n Myr, then

$$\text{rate of evolution} = (\ln x_2 - \ln x_1)/n \text{ darwins.} \quad (14.1)$$

Table 14.3 lists some results. The most obvious conclusion is that the shorter the time period over which the rate is measured, the higher the rate. This is in part an

Table 14.3

Measurement of the rate evolution (Gingerich 1983)

	Number of cases	Time-interval (geometric mean)	Rate in darwins (geometric mean)	Regression of In (rate in darwins) vs. In (time in Myr)
Selection experiments	8	3.7 yr	58 700	-0.9
Recent colonizations	104	170 yr	370	-2.08
Post-Pleistocene mammals	46	8200 yr	3.7	-0.68
Fossil vertebrates	228	1.6 Myr	0.08	-0.82
Fossil invertebrates	135	7.9 Myr	0.07	-0.59
Combined data	521			-0.85

artefact of the way in which the data were collected. Thus laboratory populations exposed to artificial selection are recorded, but not laboratory stocks maintained without selection. In the examples of recent colonization, the populations were in a new environment, and therefore probably exposed to directional selection. Examples of rapid evolution over periods of millions, rather than tens or hundreds, of years may be missing because, if they occurred, the initial and final populations might not be recognized as belonging to the same lineage.

Another possible reason for the inverse relation between time period and rate is that large changes in one trait may require that other genetically independent changes also occur. For example, it would probably be rather easy to lengthen the neck of an antelope by, say, 10 per cent. But to produce a functional giraffe by selection would require major changes in the heart and vascular system: otherwise, the animal would faint every time it lowered its head to drink. More generally, the very slow rate of evolutionary change, relative to change under artificial selection, may in part be due to the fact that, in nature, many different traits are changing simultaneously.

However, a major reason for the data of Table 14.3, and in particular for the negative relation between period and rate within categories (e.g. within invertebrate fossil lineages) as well as between categories, is that the direction of evolution is often reversed. As shown in Fig. 14.3, if there are reversals in direction, there will be a negative correlation between rate and period.

G.G. Simpson (1944 and later), one of the architects of the Modern Synthesis, emphasized how variable is the rate of evolution. More recently, critics of the Modern Synthesis (in particular, Eldredge and Gould 1972; Stanley 1979) have argued that the pattern of change usually observed is one of long periods of little or no change (stasis) interrupted by periods of rapid change (punctuation) occurring at times of lineage splitting (that is, of speciation). They have further

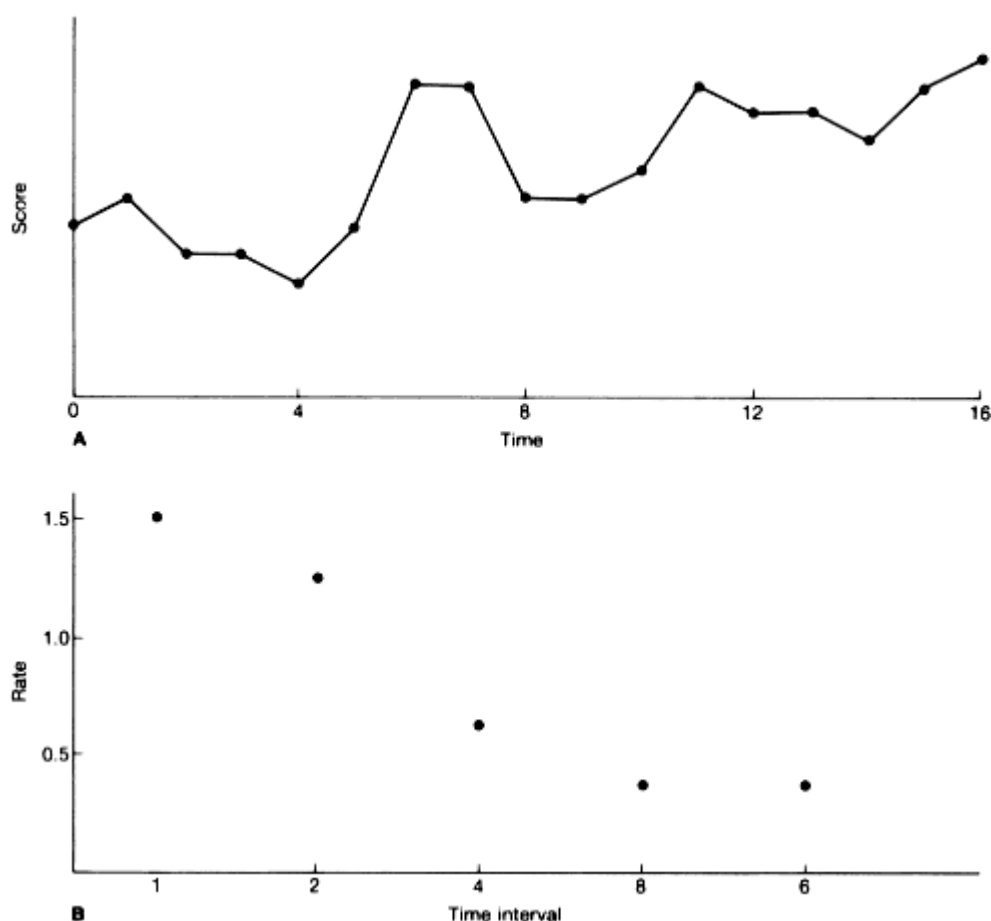


Figure 14.3

The transmission of parasites. **A**, Direct; **B**, **C**, horizontal. In **B** a host is infected from only a single source, and **C** from two sources.

argued that this pattern of stasis and punctuation means that the large-scale features of evolution (macroevolution) cannot be interpreted as the summed effects of changes of the kind that can be observed today within populations (microevolution).

The nature of the evidence on which this conclusion rests is best understood from the work of Eldredge (1985), one of the proponents of the theory, on mid-Devonian trilobites. During the Devonian, for some 8 million years, an inland sea covered much of what is now the central USA. Fossils from the mid-west, from the western shore of this sea, include three species of the trilobite genus *Phacops*. These differ in the number of rows of lenses in the compound eyes. The first, *P. milleri*, has 18 rows; the second, *P. rana*, has 17 rows; the latest, *P. norwoodensis*, has 15 rows. These species replace one another suddenly, without intermediates. The

same three species, in the same sequence, are found in New York state, from what was the eastern shore of the inland sea. However, in one quarry near Syracuse, Eldredge found a population intermediate between *P. milleri* and *P. rana*: some 50 specimens include ones with 18 rows, with 17 rows, and with 17 rows plus a variable number of lenses making up part of an 18th row. This intermediate population is some 3 million years older than the transition between *P. milleri* and *P. rana* in the west. A population intermediate between *P. rana* and *P. norwoodensis* has also been found in the east, with individuals having 15 1/2, 16, and 16 1/2 rows of lenses, but in this case it is impossible to say that this population is earlier than the sudden transition in the west, although on any rational interpretation of the data it must be.

The interpretation of these facts seems clear. On the eastern shore, the three species are connected by evolutionary intermediates. There are periods of stasis, separated by two periods of rapid evolution. On the western shore, the apparently instantaneous transitions represent the replacement of an earlier species by a later one that evolved elsewhere, in the east. Sudden replacements must often be explicable in this way. A palaeontologist of the future, studying sedimentary rocks laid down in Britain today, will observe the instantaneous replacement of the red squirrel *Sciurus vulgaris*, by the grey, *S. carolinensis*. Occasionally, we are fortunate enough, as Eldredge was with his trilobites, to observe a rapid evolutionary transition. In such cases, the rate of evolution is indeed rapid in comparison with what has gone before, and what will follow, but they are still slow compared to the rates to be expected under strong directional selection. It is therefore not obvious that the pattern of punctuation and stasis requires mechanisms other than natural selection within populations. I will return to this question on p. 281.

The pattern of stasis and punctuation may occur, but it is not a universal feature of the fossil record. Sheldon's (1987) study of trilobites during 3 million years of the Ordovician illustrates a different pattern. These animals were classified into species by the number of 'pygidial ribs' 桎 feature of the exoskeleton. In one genus, *Ogygiocarella*, the specimens available before Sheldon's study fell into two 'species', an earlier one with 11 ribs and a later one with 13. Figure 14.4 shows these data, together with Sheldon's own collections. The picture of distinct species, with few intermediates, is seen to be an artefact of the incompleteness of the record. In fact, all eight genera studied showed an increase in rib number during the period, but in no case is there a punctuational event separating an early and a later form.

It remains to be seen how common the punctuational pattern will prove to be. What is clear is that the matter can be decided only by a statistical study of populations.

Trends

It is common to observe in the fossil record a long-continued tendency to change in a given direction. For example, many lineages, both vertebrate and invertebrate,

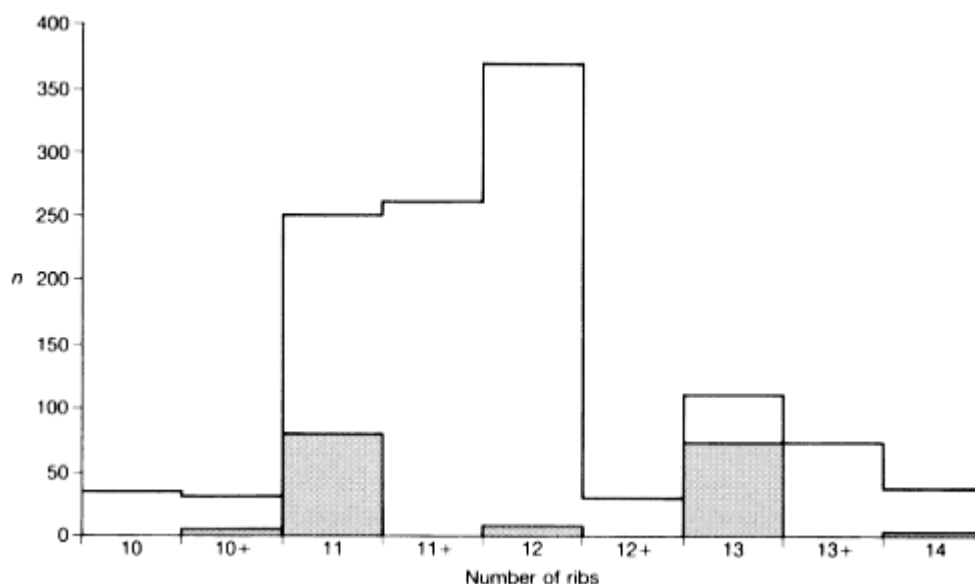


Figure 14.4

Pygidial ribs in trilobites. Intermediate numbers such as 11 + indicate 11 full ribs and one partially developed rib. The black histogram gives the numbers of individuals upon which two 'species', *Ogygiocarella debuchii* and *O. angustissima*, from different time horizons, were erected. The white histogram represents data collected later, mainly from rocks of intermediate age. (After Sheldon 1987.)

show a trend towards greater size. Herbivorous mammals usually show a trend towards hypsodonty (teeth that are high relative to their horizontal dimensions). The phenomenon is more striking when similar changes occur in parallel in related lineages. For example, many lineages of mammal-like reptiles during the Mesozoic showed parallel trends in the skull and teeth, including differentiation along the tooth row, reduced tooth replacement, development of a secondary palate, erosion of the dermal roof of the skull, and growth of the dentary relative to other bones in the lower jaw. It is hard not to interpret these changes as adaptations for chewing, or to conceive of any process other than natural selection within populations that could bring them about. During the same period, changes occurred in the limbs, girdles, and backbone, associated with the evolution of the mammalian mode of locomotion, in which the backbone is arched in a vertical rather than a horizontal plane.

Some trends can be explained in adaptive terms. Others, like the increase in the number of ribs in trilobites described above, are hard to interpret, because we know too little about the biology of now-extinct organisms. A few have been interpreted as maladaptive, and as caused by some inner and unexplained evolutionary inertia. A classic example is the great size of the antlers in the Irish elk, which may well have been a real handicap, both in escaping from predators, and as a drain on resources, particularly of calcium. However, as shown in Chapters 7 and 13, traits

can be exaggerated to a maladaptive level by intra-specific competition, and particularly by sexual selection, either by male-male competition or by a runaway process involving female choice.

Adaptive Radiation

In the early Palaeocene, the placental mammals were a rather uniform group, small and mainly insectivorous. They rapidly radiated into many adaptive zones, as herbivores and carnivores, arboreal and fossorial, in the air and in the sea. Such adaptive radiations are a common feature of the fossil record, at different scales. For example, an early group of placentals, the creodonts, evolved a range of carnivorous forms similar in morphology to modern weasels, dogs, bears, and hyenas. They were later replaced by modern fissipede carnivores, with a similar range of types, but these were a second radiation descended from a single creodont group.

An interesting but often unanswered question in such cases is this: does the newly radiating taxon cause the extinction, through competitive displacement, of the taxa earlier occupying the same ecological niches, or is the extinction of the earlier taxa the primary event, leaving the field open for the later radiation?

The analysis of such cases is not easy. Consider, for example, Fig. 14.5. One is

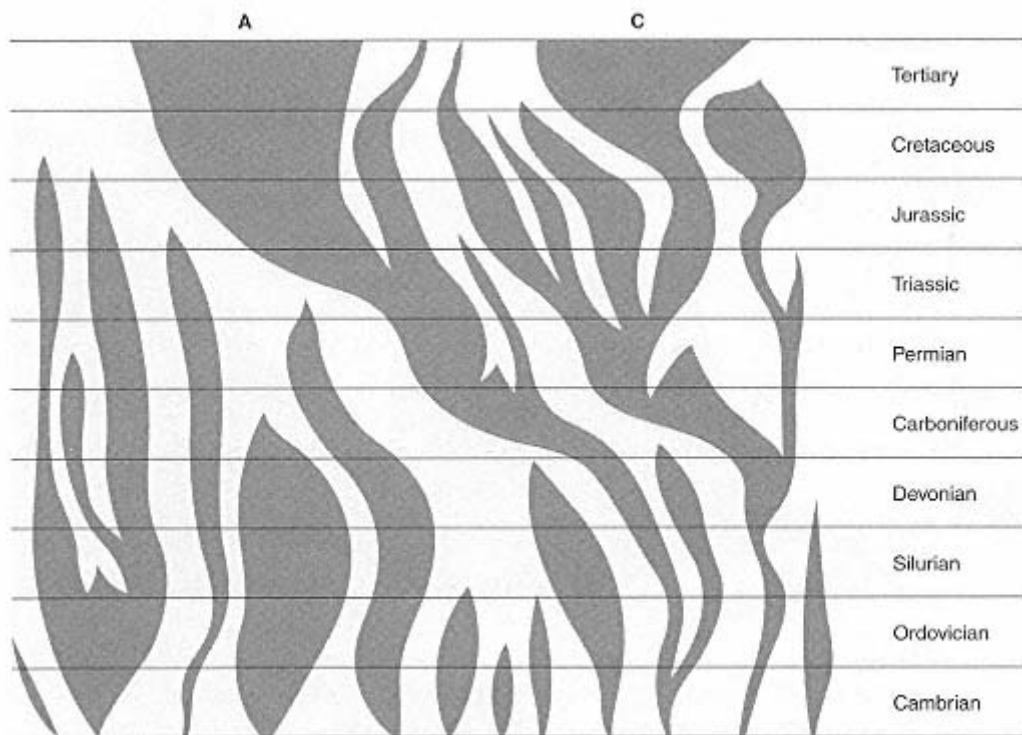


Figure 14.5

The abundance of different taxa in different geological periods: the horizontal width of a taxon at any time is approximately proportional to the number of genera.

tempted to ask, what biological invention by taxon A enabled it to radiate so successfully? What characteristic of the apparently conservative lineage, B, enabled it to radiate during the Cretaceous, and why was it later displaced by taxon C? The temptation should, however, be resisted, because the figure was drawn to represent a computer-generated phylogeny, in which the splitting and extinction of genera was a random event. Several people have found that such stochastic phylogenies have an embarrassing similarity to actual ones. The figure is not intended to demonstrate that phenomena such as adaptive radiation are not real, but only that one must proceed with caution when analysing real data.

Extinction

Figure 14.6 shows the numbers of families of marine animals present at various times in the geological record. One dramatic feature is the occurrence of several massive extinctions, the most recent being at the end of the Permian, and at the end of the Cretaceous. There is little doubt about the reality of these events, but there is

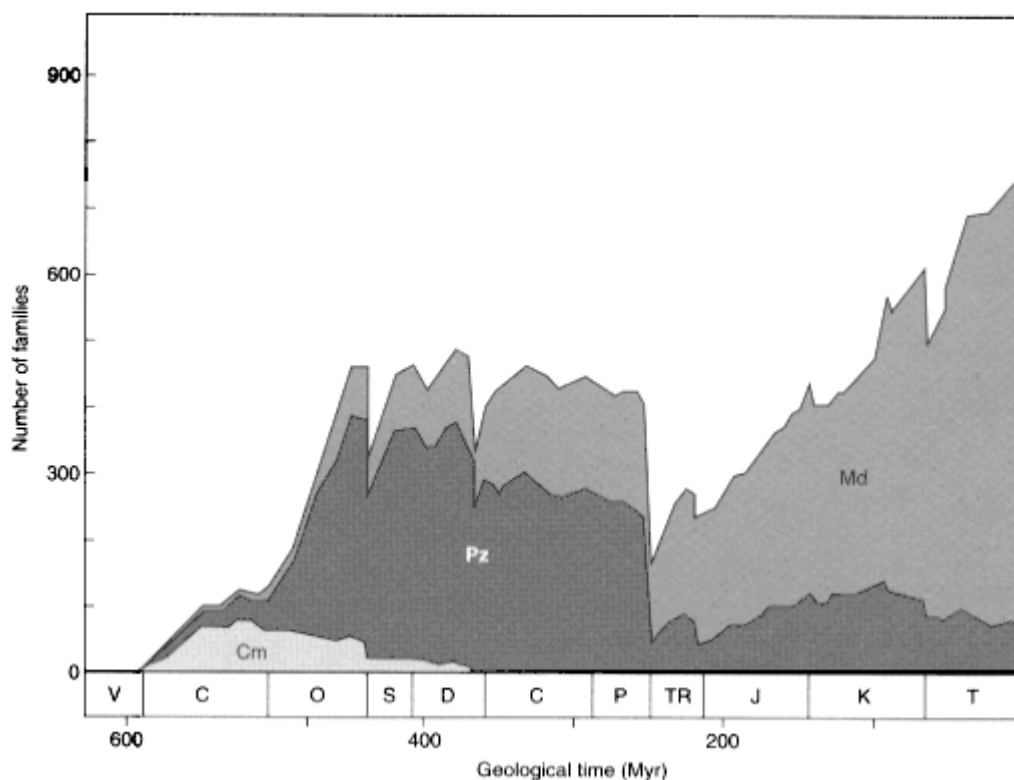


Figure 14.6

The history of taxonomic diversity of families of marine animals. The total number of families is divided into three evolutionary faunas: Cm, Cambrian fauna; Pz, Palaeozoic fauna; Md, modern fauna. (After Sepkoski 1984.)

still debate about how instantaneous were the extinctions (are we looking at something that happened overnight, or that was spread over, say, a million years?); about whether they were caused by a change in physical conditions or by some biological event (remember that the latest massive extinction ^{the} one that is happening now ^{cannot} be blamed on physical events); and about whether, if the cause was physical, it was extraterrestrial (for example, a shower of meteorites) or terrestrial (arising, perhaps, from continental drift). There is also debate about whether, in addition to these major events, there has been a series of smaller but still massive extinctions, and, if so, whether they have been irregular in time, or regularly periodic.

These mass extinctions raise further questions. Is there selectivity in which species go extinct? If so, what characteristics predispose species to extinction? After a mass extinction, is there evidence of particularly rapid speciation and diversification of the survivors?

Palaeontology and Population Genetics

It is perhaps inevitable that practitioners of these two branches of science should often misunderstand one another. It is not uncommon for palaeontologists to assert that population genetics cannot account for the fossil record, whereas population geneticists hold that there is nothing in that record that they cannot explain. Much misunderstanding would be avoided if it were understood that there is a sense in which both these views are correct. To a geneticist, there is nothing in observations such as those of Eldredge on trilobites which present any particular difficulty. Stasis can be explained by normalizing selection for a fixed optimum, and punctuation by periods of directional selection. The rates of change, although rapid by palaeontological standards, are slow to a geneticist: as J.S. Jones remarked, 'one man's punctuation is another man's gradualism'. There is no reason to suppose that the transitions took place in particularly small populations: if they had, we surely would not have been lucky enough to find traces of them. The transitions did not always, or even usually, involve the gradual transformation of the whole species, throughout its geographical range, but it has never been part of the Modern Synthesis to suppose that they did.

But such a reply, quite properly, would not satisfy a palaeontologist, who wants to know why evolution shows the patterns that it does. Why should there sometimes be normalizing selection for an unchanging optimum for millions of years? What causes punctuational events? The answers to these questions does not lie within population genetics, which only makes statements of the form 'IF there is selection of a given kind, and IF a population has certain properties, THEN the following changes will occur'. But genetics cannot tell us what the nature of the selective forces will be. It is in this sense that I think palaeontologists are right to say that population genetics cannot account for evolution.

If we are to gain an insight into the nature of selective forces, it must come, I

think, from a study of ecology. In particular, it must come from a study of the coevolution of interacting species, because the main selective forces acting on a species are likely to come from changes in its competitors, its predators, and its parasites. This is the topic of the final section in this book.

This, however, is not the conclusion drawn by many palaeontologists, who seek to explain the major patterns of evolution, not in terms of the selective forces acting on populations, but in terms of developmental constraints that limit the ways in which a population can respond to selection. There are such constraints. For example, monocotyledonous trees (palms) do not branch, unlike characteristic dicots such as oaks, maples, and poplars. The reason is that monocots never evolved a process of secondary thickening, whereby a twig can turn into a branch. In other words, a developmental constraint (absence of secondary thickening) has limited the evolution of a major taxon. As it happens, the constraint is not absolute: some monocots do branch (see Fig. 14.7).

The range of variation possible to a taxon, then, is limited by the pattern of development that has evolved in that taxon. Occasionally, as in the example of palm trees, we have some idea of the reason for the constraint. More often, we do not. To take another example from the monocots and dicots, the former almost

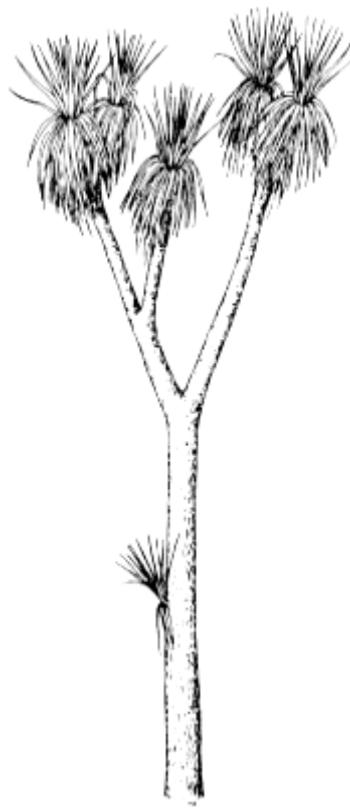


Figure 14.7
Cordyline australis, a
branching monocot.

always have flowers with three petals and three sepals: so far as I know, no monocot has a typical flower with a petal number greater than three. This cannot be explained in selective terms, because dicots, faced with the same range of potential pollinators, typically have four, five, or more petals. It is natural to suspect a developmental constraint, but we do not know what it is. It is not an absolute one, because occasional flowers with four sepals, petals, and stamens do occur (Fig. 14.8). Also, the constraint has not prevented the monocots from repeatedly evolving bilaterally symmetrical flowers: orchids are the most dramatic example.

The evolution of a taxon, then, depends on selective forces, which arise mainly from ecology, and on the available range of variation, as determined by its mode of development, which in turn reflects the past evolution of the taxon. Does the study of the fossil record suggest any mechanism of evolutionary change other than

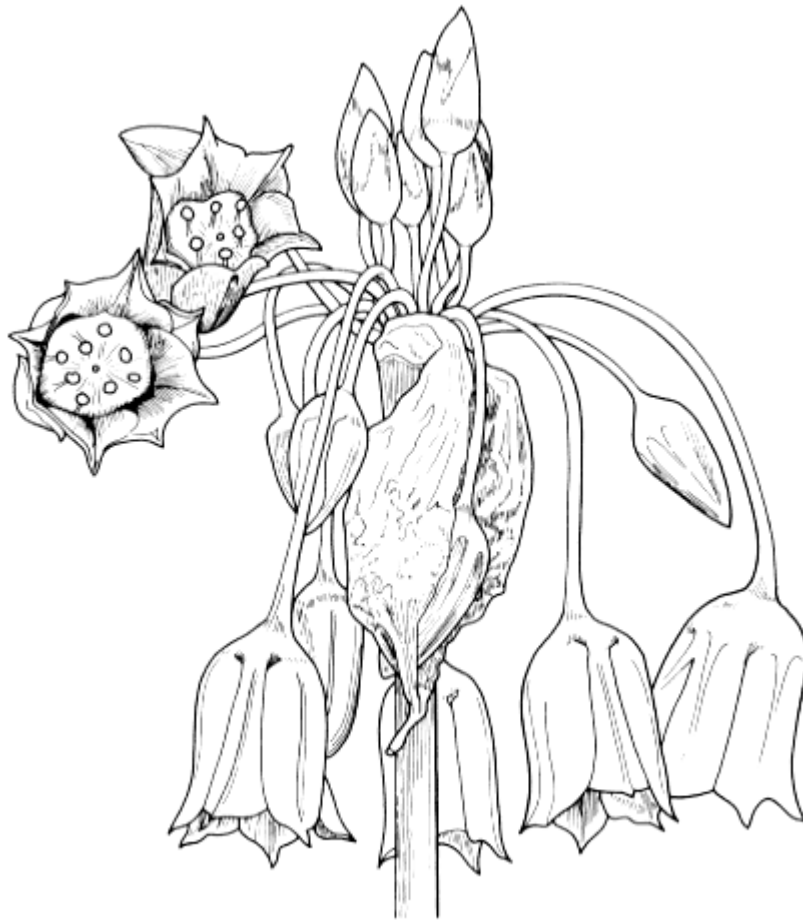


Figure 14.8

Flower head of *Allium siculum*, showing typical flowers with three sepals, three petals, and six stamens, and an atypical flower with four sepals, four petals, and eight stamens.

natural selection acting on the available variation, together, of course, with stochastic changes in gene frequency? A number of suggestions have been made, but most are not needed to explain the observed phenomena, and assume the existence of processes for which there is no basis in what we know of biology. One process, species selection, does need more serious discussion.

In effect, the suggestion is that the units of evolution are the entities that display multiplication, variation, and heredity, and which therefore evolve by natural selection, are species rather than (or as well as) individuals. In this model, births are replaced by speciation events, deaths by extinctions, and mutations by the punctuational changes that occur when lineages split. The punctuational theory, which in its extreme form holds that the transitions between species are effectively instantaneous and non-adaptive, leads necessarily to the view that species selection is the major directing force in evolution: there is, of course, no good reason for accepting this extreme view.

It is, I think, important to distinguish two processes, only one of which should properly be called species selection:

1. The replacement of species A by species B because of the superior competitive ability of B individuals. For example, consider the replacement of creodonts by modern carnivores, mentioned above. If (and this is an assumption) a particular creodont species was replaced by a fissipede species that was competitively superior perhaps because it was better at hunting the same prey then the process was essentially one of individual selection. Individuals competed for the same limiting resource, and the fact that they belonged to two different species has little to do with it. The traits by virtue of which the fissipedes won presumably evolved by natural selection between individuals in an ancestral fissipede population.
2. An increase in the number of species in taxon A, relative to the number in taxon B, because some characteristic of the species in taxon A make them more likely to speciate (equivalent to an increased birth rate), or less likely to go extinct (a reduced death rate). One example of this process was discussed in Chapter 12, where I argued that one reason for the maintenance of sexual reproduction is the lower capacity for evolution of asexual taxa. Two points are worth making about this example. First, the relevant trait, capacity to evolve, is characteristic of the species, or population, and not of the individual: populations, and not individuals, evolve. Secondly, it is unusual for there to be within-population variability for mode of reproduction, sexual or asexual: it is this that makes the species-selection explanation plausible. The fact that there is sometimes within-population variation means that species-selection cannot be the whole explanation: this is Williams' (1975) 'balance' argument.

A second possible example has been discussed by Vrba (1984). It concerns two sub-families of African antelopes, during the past 5 million years. The

Aepycerotini has, over the whole period, consisted of a single species, the impala, which feeds on a wide range of plants, and does not migrate. A second family, the Alcelaphini, are food specialists, and are therefore obliged to migrate to follow the available food. During the same period, there has been repeated speciation and extinction. In all, 28 species have been recognized, of which seven survive today. It seems that the main cause of the different patterns of evolution is the greater speciation rate of the Alcelaphini. If so, the important cause of speciation has probably been the ecological fact of food specialization, and hence the existence of unoccupied ecological niches, and not population structure: if the latter had been the important factor, we would expect the non-migratory impala to speciate more rapidly, because of the greater opportunities for geographical isolation. In any case, if we measure success by number of species, the Alcelaphini have increased sevenfold in 5 million years, and are out-competing the Aepycerotini. But it is not clear that this is the right measure of success: as Vrba points out, there are probably more individual impala in Africa than there are of all the Alcelaphini species together.

How, then, should we evaluate species selection? It is a logically possible process, which may have influenced the evolution of breeding systems, patterns of dispersal, and degree of ecological specialization, but is largely irrelevant to most of the morphological characteristics that are the subject matter of palaeontology.

Coevolution

If a trait of one species, A, has evolved because of the presence of a second species, B, and if a trait of species B has evolved because of the presence of A, then **coevolution** has occurred. In discussing coevolution, it is convenient to distinguish three cases, according to the nature of the ecological relations between the interacting species:

1. **Competition:** the presence of each species inhibits the population growth of the other.
2. **Exploitation:** the presence of species A stimulates the growth of B, and the presence of species B inhibits the growth of A. Examples are plant-herbivore, prey-predator, and host-parasite interactions.
3. **Mutualism:** the presence of each species stimulates the growth of the other.

Competition

If two species compete for, and are limited by, a single resource, then theory indicates that one or other will be eliminated (p. 21). If, however, they compete for a range of resources, it is possible that they will partition the resources between them, and that both will survive. Theoretical models of resource competition suggest that two similar species will usually evolve so as to take different resources.

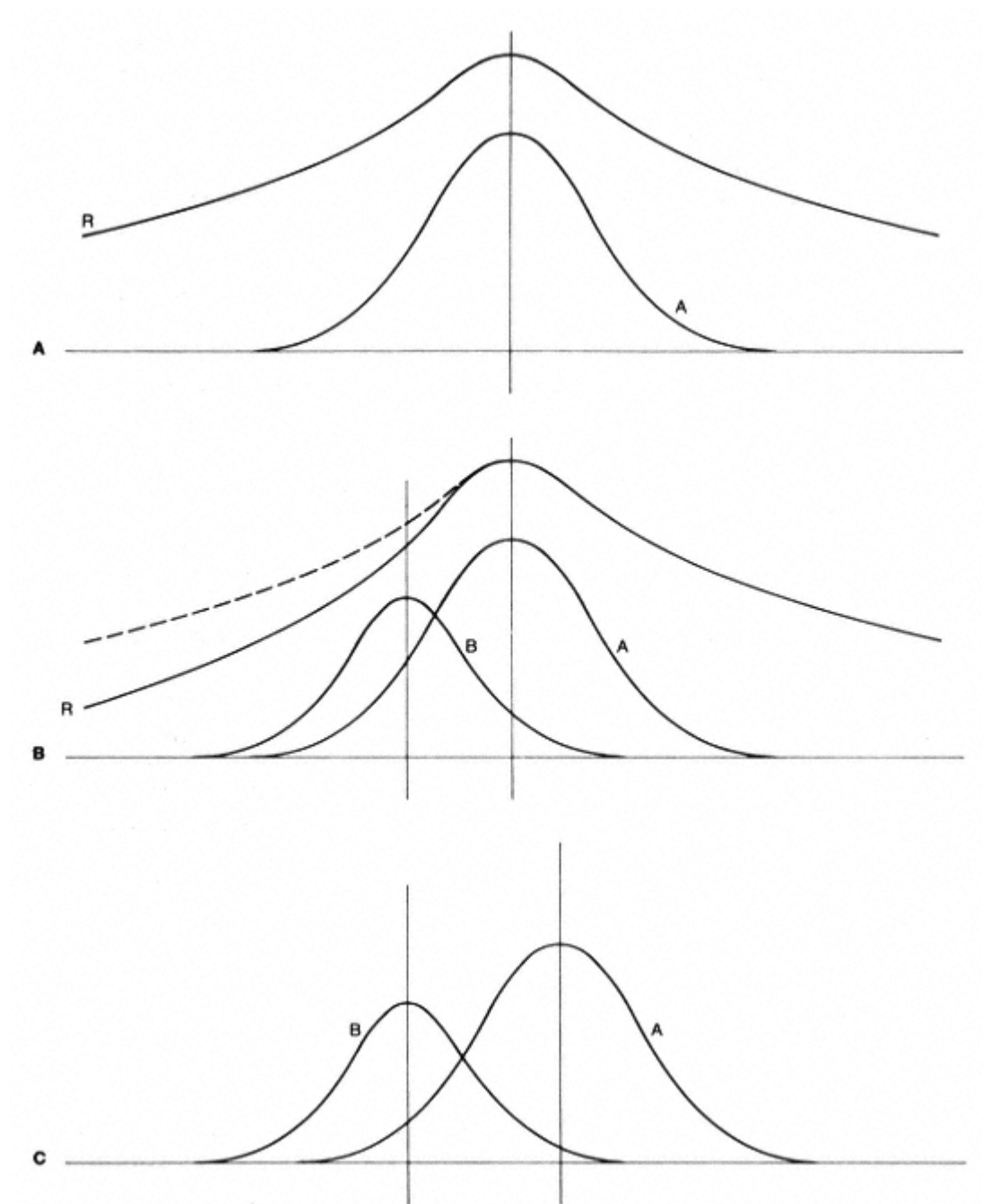


Figure 14.9

A model of character displacement. **A** Resource utilization of species A on its own, given available resources R. **B** Change in available resources, R, induced by the presence of species B; the original resource distribution is shown by the broken line. **C** New resource utilization by species A.

If this results in the two species becoming morphologically more different, the process is referred to as **character displacement**: note, however, that character displacement can occur for reasons other than resource competition. For example, selection for mating isolation.

A simple argument leading to this conclusion is illustrated in Fig. 14.9. Suppose that species A has, on its own, evolved an optimal phenotype for exploiting a given range of resources. The population will vary genetically, as all populations do, but individuals above and below the mean will have equal fitnesses: the population is under normalizing selection. Now suppose that species B, with an initially different resource utilization (below the mean of A in the illustration), is introduced. The effect will be to lower the resource availability, and hence the fitness, of A individuals below the mean. Hence species A will evolve away from species B: that is, character displacement will occur. Of course, the displacement will continue only until A has reached a new optimum, for the new range of resources that are available when species B is also present. For a more rigorous treatment of this problem, see Roughgarden (in Futuyma and Slatkin 1983).

Note that an essential assumption is that competing species have a significant impact on the availability of the resource. If, for example, species A and B were limited by their predators or parasites to a low population level, and so had no effect on the availability of their food, we would not expect to observe character displacement.

One of the clearest examples of character displacement concerns two species of marine mud snails, *Hydrobia ulvae* and *H. ventrosa*, in a channel, the Limfjord, in northern Denmark. Although the species have different salt tolerances, their ranges overlap. They feed by ingesting the substrate, and digesting the microorganisms attached to the mineral and detrital particles. Figure 14.10 shows that, when only one species is present, the shell size is the same in different populations. But, if both species are present, *H. ulvae* is larger than *H. ventrosa*. Also, in regions of overlap, the difference in body size is associated with a difference in the size of the ingested particles.

Exploitation.

Plant-Herbivore Interactions

Much of the information we have about the coevolution of plants and herbivores comes from a study of phytophagous insects. When a plant species is introduced into a new region, the herbivorous insects that attack it are usually local in origin. Some introduced plants are so different from the local flora that no native insects are able to attack them. For example, eucalyptus trees are virtually free of insect attack in California, although many insects feed on them in their native Australia. The same immunity is enjoyed by the prickly pear, *Opuntia*, introduced from America into Africa. More commonly, introduced plants do suffer from insect

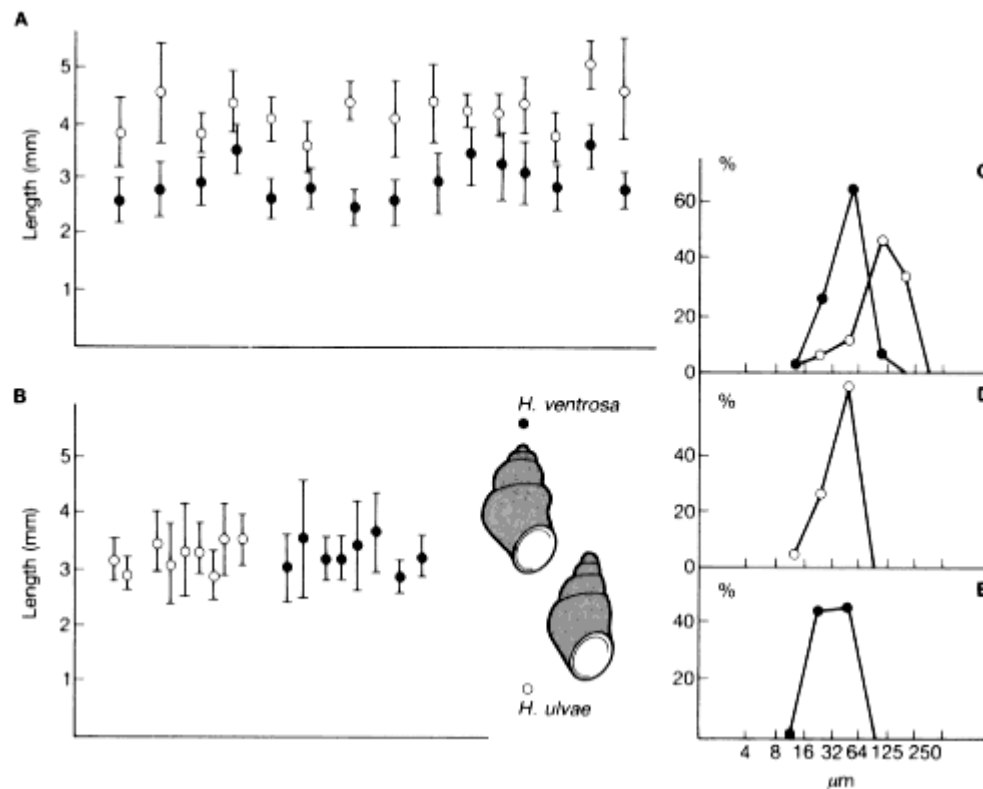


Figure 14.10

Shell lengths of two species of mud snails in the Limfjord, northern Denmark: **A** 15 localities with coexistence; **B** 17 localities with one species only; **C** size distribution of ingested food particles in coexisting populations; **D, E** size distribution of food particles in allopatric populations. (After Fenchel and Christiansen 1977.)

attack. Sugar cane, probably a native of New Guinea, has been introduced into many countries, the earliest being India in 1000 BC, and the most recent Honduras in 1840. The number of local insects that have colonized sugar cane in any region increases with the area over which the species is cultivated. Perhaps surprisingly, it does not correlate with the length of time since its introduction. This suggests that, in any locality, there is a limited number of insects capable of colonizing sugar cane, and that these move to the new host fairly quickly. The number of potential colonists will be greater if the region sampled is large, and if the introduced plant is not too different from plants already present. An example of the way in which a local insect may be pre-adapted to an introduced plant is the fact that the cabbage white butterflies of western Europe, *Pieris brassicae* and *P. rapae*, now attack the introduced garden nasturtium, which is taxonomically unrelated to the Cruciferae that are their natural hosts, but which shares with them the presence of mustard oils as a defence against phytophagy.

These facts suggest that introduced plants are colonized only by those insect species in some way pre-adapted to do so. It may be that, over a longer time-period, additional species would make the switch, but there is evidence that particular feeding niches on plants may remain unoccupied for long periods. Figure 14.11 shows the insect species feeding on bracken, *Pteridium aquilinum*, at three intensively studied sites at which bracken is native. These data strongly suggest that there are, at each site, unoccupied ecological niches, and hence that there are no local insects capable of making the switch.

The structural and chemical defences of plants have presumably evolved by

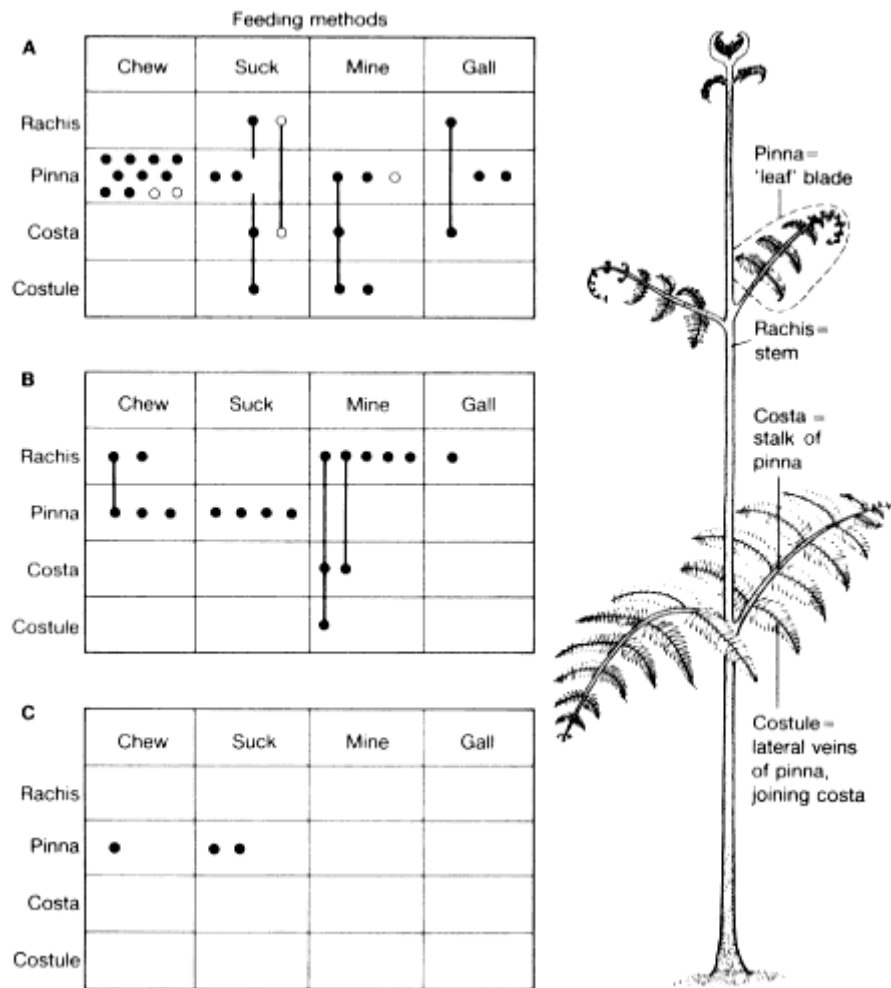


Figure 14.11
Feeding sites and feeding methods of insects attacking bracken. **A** Skipwith Common, Yorkshire, England; **B** Papua New Guinea; **C** Sierra Blanca, New Mexico. None of the sites have any insect species in common. Each dot represents one species; feeding sites of species exploiting more than one part of the frond are joined by lines. (After Strong *et al.* 1984.)

Table 4.4

Degree of host-specilization in insects collected in Umbelliferae differing in chemical composition (from Berenbaum 1981, quoted in Strong *et al.* 1984)

Host plant chemistry more	Proportion of		
	Extreme specialists, feeding on only 1-3 genera	Intermediate species, feeding on 4-20 genera	Polyphages, feeding on than three families
With angular and linear furanocoumarins	0.43	0.285	0.285
With linear furanocoumarins only	0.30	0.30	0.40
Without furanocoumarins	0.00	0.36	0.64

natural selection, but the facts so far quoted show little evidence of coevolution. The evidence is more consistent with the view that a local insect is either already capable of switching to the introduced plant, or it is not. More positive evidence of coevolution is given in Table 14.4. The data indicate that some Umbelliferae have evolved particularly toxic compounds (furanocoumarins), and that, in response, a few species of insects (for example, caterpillars of the moth family Oecophoridae and of the butterfly genus *Papilio*) have evolved the ability to cope with the toxins. This is evidence of the kind of arms race one would expect. Some of the ways in which insects overcome the toxins are remarkable. In direct sunlight, furanocoumarins cross-link and inactivate DNA; some caterpillars protect themselves by feeding within a rolled-up leaf.

One theoretical point is worth emphasizing. There is evidence that, at least in many cases, herbivores are limited by their predators and parasites, rather than by their food. However, it does not follow that herbivores are not an important selective force influencing the evolution of plants. A factor can be selectively significant without being density limiting.

Prey-Predator Interactions

The only attempt known to me to follow coevolution in the fossil record is Bakker's (in Futuyma and Slatkin 1983) study of mammalian herbivores and carnivores, and in particular of open-country species that escape, or catch their prey, by running. In many independent lines of evolution, both of herbivores and carnivores, high speed is achieved by the same morphological changes: in particular, by lengthening the distal relative to the proximal elements of the limb (that is, the foot relative to the femur), by straightening the limb, by reducing the number of elements in the foot from the primitive number of five, and by modifying the joints so as to confine movement to a single plane. Figure 14.12

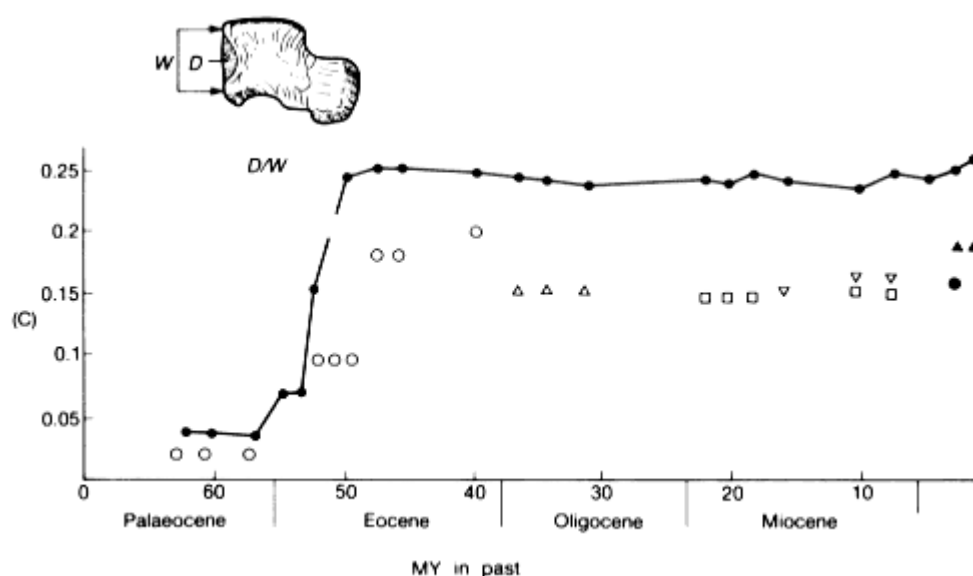


Figure 14.12

Coevolution of carnivores and ungulates. The index D/W measures the depth of the groove in the astragalus, which indicates the degree to which the limb was constrained to move in a single plane. The full line represents a series of ungulates: they are not a phylogenetic series. The single points are carnivores: ○, mesonychids; △, hyaenodontids; □, amphicyonids; ▽, borophagines; ●, hyaenids; and ▲, canines. (After Bakker, in Futuyma and Slatkin 1983.)

shows the progress of these changes, for one trait, in ungulates (which do not form a phylogenetic series), and in several carnivore lineages: other traits show a rather similar pattern. Bakker interprets these data as showing that the carnivores lagged behind the herbivores in the evolutionary race. He explains this in 'species selection' terms. The carnivores evolved more slowly because there were fewer species, and so fewer opportunities for species selection.

I do not think the data support this interpretation. They suggest that a sequence of separate carnivore lineages each evolved very rapidly, but to a degree of specialization less than that in herbivores, and then ceased to evolve. (It is not clear why, successively, one carnivore taxon went extinct, and was replaced by another.) The question to answer, therefore, is why did the carnivores not evolve such highly specialized limbs, and not why they evolved so slowly. A plausible answer is that carnivores use their limbs for things other than running (in particular, for digging lairs), whereas most ungulates do not. It is, in any case, relevant that a carnivore with a lesser degree of limb specialization can run as fast as a herbivore with a greater degree. One reason for this is that carnivores do not run in the same way as herbivores: they retain a more flexible backbone, and by arching their backbone they can lengthen their stride without lengthening their limbs.

The data that Bakker has summarized are fascinating. They suggest that there

has been an arms race, but one in which, for long periods, neither side was able to make further improvement. I do not think they support a species selection interpretation.

Host-Parasite Interactions

In Chapter 10, it was suggested that an arms race between parasites and their hosts is a major reason why populations are under continuing selection for change, and that this may in turn explain the superiority of sexual over asexual populations. There is, however, an older but still popular view that, in the words of Dubos (1965), 'Given enough time a state of peaceful coexistence eventually becomes established between any host and parasite'. In the only example of host-parasite coevolution so far discussed—the that of the myxoma virus and its rabbit host—there is clear evidence that the parasite has evolved so as to become less virulent, although the situation is still far from one of 'peaceful coexistence'. Can we make any general prediction about whether we would expect a host and symbiont to evolve towards parasitism or mutualism?

One factor which we would certainly expect to affect the outcome is the mode of transmission (Fig. 14.13). If the symbiont is transmitted only to the offspring of the individual host it inhabits—**direct** transmission; Fig. 14.13A), we would expect evolution towards mutualism, because a parasite that reduces the fitness of its host also reduces its own fitness: there is no advantage to killing the goose that lays the golden eggs. In contrast, if transmission is **horizontal** (Fig. 14.13B, C), there may be selection for increased virulence, if greater virulence is associated with greater likelihood of transmission. If a host is typically infected by only a single parasite strain (Fig. 14.13B), then, as May and Anderson (1983) argued in their analysis of the myxoma virus, selection will favour parasite genotypes that produce an optimum compromise between a high rate of infection and keeping the host alive. If, on the other hand, a host is typically infected by several parasite strains (Fig. 14.13C), selection on the parasite will favour high infectivity at the expense of host survival: there is no point in keeping the goose alive if someone else is going to kill it.

Two examples will illustrate the relationship between virulence and mode of transmission. The first concerns the nematodes that attack figwasps (Herre 1993). Mature female figwasps enter the fig inflorescence, lay eggs, and die. If the female is infected by nematode parasites, these emerge from her body and lay eggs in the same fig. Hence, if only a single wasp enters a fig, any nematodes that infect her will produce offspring that infect her offspring: that is, infection is direct. But if several females infect a single fig, transmission is, in part, horizontal. Herre found that, in species with only one wasp female per inflorescence, the fertility of a female was not reduced by nematode infection, whereas in species in which several wasps enter a single inflorescence, there is a substantial reduction in

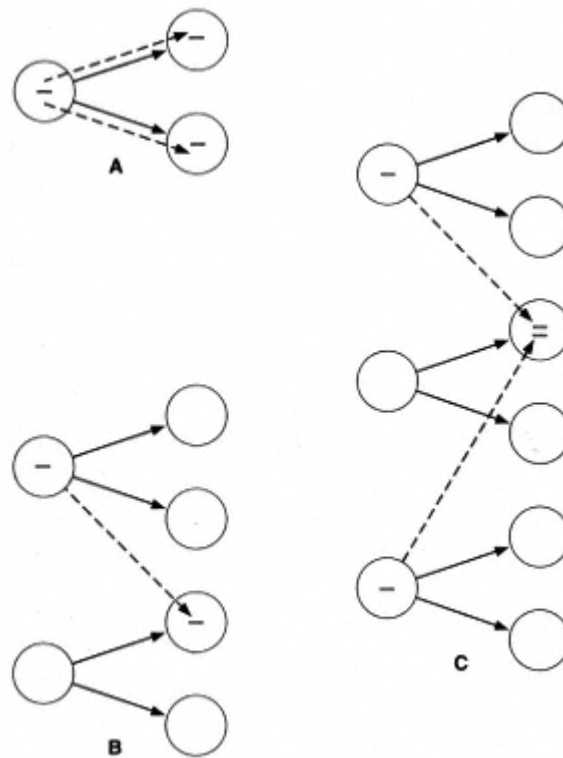


Figure 14.13
 Direct and horizontal transmission of symbionts.
A Direct transmission. **B** Horizontal transmission.
C Horizontal transmission with more than
 one infection of a single host.

fertility. In this case, it is likely that the commensal situation is ancestral, and that parasitism has evolved in species with horizontal transmission.

Evolution has taken the opposite course in the ergot-producing fungi infecting grasses (Clay 1988). There are two sexual genera, *Balamsia* and *Epichloe*, producing horizontally transmitted spores. The ergot that they produce protects the plant against herbivores, but they sterilize the host, and so must be regarded as parasites. In contrast, some ergot-producing fungi in the genus *Acremonium* are asexual: their spores are directly transmitted in the host seed, and, as predicted by the theory, they do not sterilize their host.

There is a reasonable fit to the prediction of an association between vertical transmission and a commensal or mutualistic relationship, and between horizontal transmission and parasitism, although, as described in the next section, there are puzzling examples of mutualistic relationships in which transmission is horizontal. More generally, there is much support for the view that host-parasite interactions are an important cause both of genetic polymorphism, and of continuing evolutionary change.

Mutualism

The evolution of mutualism, in which both partners benefit from the relationship, has received less attention than that of competition and exploitation, although the recognition that symbiosis played a role in the origin of eukaryotes (Margulis 1970, 1981) has stimulated interest in other examples of symbiosis. The first point is that mutualism plays an important role in many ecosystems, as the following examples will show:

1. Nitrogen fixation. Plants cannot fix atmospheric nitrogen. Leguminous plants, however, form a symbiotic relationship with the nitrogen-fixing bacterium *Rhizobium*.
2. The ecosystem of deep-sea vents depends on symbiosis. Most ecosystems depend ultimately on photosynthesis for their energy. Even deep-sea organisms, in the absence of light, depend on the fall-out of dead organisms from the surface layers, which do depend on photosynthesis. In deep-sea vents, however, the sulphides emerging from the vents provide an alternative source of energy. The vestimentifera worm, *Riftia*, that inhabits these vents has no mouth or anus as an adult, and relies on symbiotic bacteria, housed in a special organ, which oxidize the sulphides. Both oxygen and sulphur are transported to these organs by a special haemoglobin. Other invertebrates inhabiting the vents also depend on symbiotic sulphur bacteria.
3. Coral reefs are built by coelenterates containing symbiotic dinoflagellate algae.
4. Many tropical plants, growing on mineral soils, depend on symbiosis with mycorrhizal fungi. There is a net flow of minerals from fungi to plants, and of carbon from plants to fungi, so the interaction can reasonably be regarded as mutualistic. Ericaceous plants (heathers, rhododendrons), growing on acid soils, depend on a more recently evolved group of mycorrhizal fungi.

The list, which is by no means exhaustive, shows the ecological importance of symbiosis. Surprisingly, despite the prediction discussed in the last section, transmission is often horizontal. For example, vestimentiferan worms must, as larvae, swallow sulphur bacteria, and luminous fish must swallow the bacteria that make them luminous, yet in both cases survival of the host depends on the symbiont. There is, therefore, a need for further research on the maintenance of mutualism, as opposed to parasitism, in such cases.

Two facts about mutualists were discussed by Law and Lewis (1983):

1. The taxonomic diversity is low. For example, only four species of *Rhizobium*-like bacteria, belonging to two genera, have been recognized, but they form nodules in 17500 plant species, belonging to 600 genera. Only two genera of dinoflagellates are known to live within marine invertebrates, but the latter belong to three phyla 棘皮动物, Mollusca, and Platyhelminthes. This extreme non-specificity of mutualistic inhabitants contrasts with a high degree of specificity in parasites.

2. Mutualists are often asexual. The asexual ergot-producing fungus *Acremonium*, mentioned in the last section is an example. A larger-scale example is the fact that the primitive (vesicular-arbuscular) mycorrhizal fungi associated with tropical plants are all asexual.

These two features of mutualists are clearly related: both suggest that there has been little pressure for evolutionary change. This is understandable. The host species, which gains from the presence of mutualists, will evolve so as to provide suitable conditions to encourage invasion by potential mutualists present in the environment. Hence different host species will converge in this respect, although they may diversify in other ways to meet external conditions. A single inhabitant species, therefore, will be able to live in many different hosts. (For a more formal treatment, see Law and Koptur 1986.)

The origin of a new mutualism is unique in being a process whereby a single 'organism' (host plus symbiont) can acquire a large increase in adaptive genetic information in a single step. Of course, it does not remove the necessity for natural selection to generate the adaptive information in the first place, and further selection, after the origin of the mutualism, will be required to co-adapt the two, originally separate, genetic systems. Yet the possibility of combining the structural or locomotory abilities of one partner with the biochemical abilities of the other is one that has been exploited repeatedly in evolution, particularly in the colonization of new habitats.

The Red Queen

In real ecosystems, a species interacts, not with one species of competitor or parasite, but with many. As each species evolves, it alters the environment of many others, and the environment of each species alters as others evolve. This led Van Valen (1973) to propose that, even in a constant physical environment, evolutionary change will continue indefinitely, as each species evolves to meet changes in others. He called this the Red Queen hypothesis, because the Red Queen said to Alice, 'here, you see, it takes all the running you can do to keep in the same place'.

Attempts to model the coevolution of many species in an unchanging physical environment face obvious difficulties. It is hard to reach any conclusion, other than that the Red Queen picture is plausible, but not necessary (Stenseth and Maynard Smith 1984). It is quite consistent with what we know of population biology that the species in a multi-species ecosystem should continue to evolve indefinitely. Unfortunately, it is equally consistent with what we know that evolution should gradually slow down and stop, when each species has reached a local selective optimum. We are therefore faced with two possibilities.

1. In the absence of physical changes (continental drift, ice ages, meteorites), evolution would gradually slow down and stop, as the evolution of RNA

molecules in a test-tube appears to do, when each species had reached a local optimum. That evolutionary change has, in fact, continued, for the past 3 billion years, must be explained by changes in the physical environment.

2. Even in the absence of physical change, coevolution would continue indefinitely. This is not to deny that physical changes occur, and affect evolution. It merely asserts that such changes are not necessary to maintain evolution.

I do not at present see any way, from ecological and genetical theory, to choose between these alternatives. It will be difficult to choose between them from a study of the fossil record, but that may be the only way.

Further Reading

On species and speciation:

Coyne, J.A. and Orr, H.A. (1989). Patterns of speciation in *Drosophila*. *Evolution* **43**, 362-81.

Mayr, E. (1963). *Animal species and evolution*. Harvard University Press.

Templeton, A.R. (1980). The theory of speciation via the founder principle. *Genetics* **94**, 1011-38.

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On patterns of evolution 根 or two opposing views:

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On coevolution:

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Problems

1. Human brain volume has trebled in the last 4 Myr. Suppose that the heritability of brain volume is 0.5, the coefficient of variation (SD/mean) is 0.1, and the average generation time has been 15 years. What intensity of selection, I , would be needed to produce the observed change?

2. Suppose that you have estimates of the number of genera of artiodactyls, and of perissodactyls, at 10 approximately equally spaced times during the last 30 Myr. The overall trend has been an increase in the number of artiodactyl genera relative to perissodactyl. How could you test the null hypothesis that this change has been entirely a matter of chance?

3. S.M. Stanley reports that, in the history of the elephants during the past 5 Myr, three genera (*Mammuthus*, *Loxodonta*, and *Elephas*) appeared during the first 0.5 Myr, and that no further genera have appeared since that time. He interprets this as showing that evolution was rapid during the early evolution of the family, and has slowed down since.

Is there any other explanation of the observations? How could you distinguish between different explanations?

Computer Projects

1. Imaginary phylogenies. Write a program to generate data similar to those on which Fig. 14.5 was based. Assume random extinction and speciation, but a constant number of genera. Consider how you might distinguish between real data and stochastic phylogenies of this kind. One way in which real and simulated data may differ is that 'living fossils' are relatively rare in simulated data. How could you test this? If it is correct, how could it be explained? (It is not difficult to generate suitable data, but harder to find a neat way of representing the data graphically.)

Chapter 15— Reconstructing Evolutionary History

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Pre-literate people assign living organisms to natural kinds: cats, dogs, oak trees, and so on, and then group these kinds into higher taxonomic groups. Long before Darwin, this process of classification was a major preoccupation of biologists. Since Darwin, it has been increasingly accepted that classification should reflect evolutionary relationships. More precisely, a taxon should be **monophyletic**: that is, it should include all the descendants, and only the descendants, of some common ancestor. Although this is sometimes contested, it is clear that such a phylogenetic classification is the kind of classification needed by biologists.

Classification is not normally seen as a branch of genetics. However, since it relies increasingly on molecular data, it seems right to give a brief introduction here. I will concentrate on the logic of tree construction, which is similar whether one is using molecular or morphological data. There are, however, two points that need emphasizing about the use of sequence data. The first is that the choice of molecule depends on the group being classified. For a group of closely related organisms, one needs a rapidly evolving molecule that varies sufficiently within the group: for example, mitochondrial DNA. For a more distantly related group, one needs a slowly evolving molecule, such as the DNA coding for ribosomal RNA, so that resemblances between the more closely related members of the group are still recognizable. The second point concerns the likelihood of parallel changes occurring in different lineages. Morphologists can reasonably assume that the pentadactyl limb, or Aristotle's lantern of echinoderms, evolved just once. In contrast, the substitution, say, of an A by a G at a particular site in a gene could well occur independently in two different lineages. This matters, because to allow for such repeated events, or **homoplasies**, is a major difficulty in tree construction.

A necessary assumption in the construction of a phylogenetic tree is that the objects to be classified arose by a branching process. It is also usual to assume that all branches were dichotomous: that is, a single ancestral lineage split to give rise to two descendent lineages. If two such splits followed closely one after the other, it may in practice be impossible to decide which occurred first, but the (perhaps

unjustified) assumption is usually made that, if we knew enough, all splits would be seen to be dichotomous. More important, it is assumed that lineages split, but never rejoin: history is to be represented by a tree, not by a net. This assumption is not true of all objects one might wish to classify: it would not be sensible to construct a phylogenetic tree of the students attending a lecture, or of the cars in the car park. In particular, if there has been genetic recombination between lineages, a tree is an inappropriate representation.

Groups for which tree construction is justified include the following:

- (1) members of an asexual population;
- (2) members of different sexual species, provided that inter-species hybridization can be ruled out, or of higher taxa;
- (3) mitochondrial genes, either between species, or, if inheritance is strictly maternal, within species;
- (4) Y chromosomes;
- (5) chromosomal genes, provided that intragenic recombination has not occurred.

How to Construct a Phylogenetic Tree.

Given that tree construction is valid, how is it to be done? Figure 15.1 gives imaginary data for four traits, *A-D*, in four individuals, 1-4. The traits could, in principle, be morphological or molecular, but for the moment suppose that they represent synonymous substitutions in four codons within a gene. Figure 15.1B is a possible phylogenetic tree, generating the four genotypes from a common ancestor. Note the following points:

1. The tree is a **maximum parsimony** tree: that is, a tree containing the minimum number of steps, or changes, needed to generate the observations. For larger data sets, it is often the case that a number of alternative, equally parsimonious trees exist.
2. The tree contains five steps, although there are only four polymorphic sites. One site *A*, has changed twice: as explained above, such a repeated event is called a homoplasy.
3. At one site, one alternative allele (*B*) is present in only one individual. Such a site is said to be 'uninformative', because it contributes nothing to finding the most parsimonious tree: whatever tree is chosen, variation at an uninformative site can always be explained by a single event in the terminal branch leading to the unique individual.
4. The tree has been 'rooted' in a common ancestor. This is not justified by the data, which are explained equally well by the tree shown in Fig. 15.1B. In fact, the data only justify the 'unrooted' tree shown in Fig. 15.1C.

In practice, trees are usually constructed using one of many computer packages, each offering a number of methods. How should we choose an appropriate

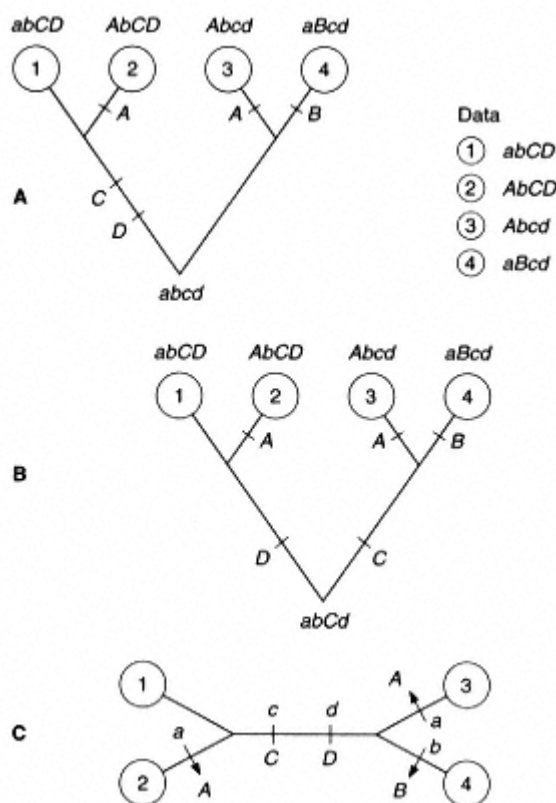


Figure 15.1
Constructing a phylogenetic tree. **A**, data, and a possible tree; **B**, an alternative tree with a different root; **C**, an unrooted tree, which is all that the data justify.

method? It seems natural to seek a tree that satisfies some 'optimization criterion'. A maximum parsimony tree does this. It seems sensible that a tree requiring a small number of evolutionary changes is more likely to be right than one requiring many changes. However, this need not be the case. Thus suppose that change *A*, which occurs twice in the tree in Fig. 15.1**A**, was not a single base change, but an intragenic inversion, with precisely the same break points. It is very unlikely that such a change could have occurred twice. If we insist that change *A* was unique, we are led to the tree in Fig. 15.2, which has six steps.

A second reason why maximum parsimony may give the wrong answer is illustrated in Fig. 15.3. The essential feature is that the tree has very different branch lengths. Imagine that all changes are synonymous transitions ($A \rightleftharpoons G$ or $T \rightleftharpoons C$). Eight typical sites, *a-h* are illustrated. Comparing taxa 3 and 4, there are three kinds of site:

1. Sites *a* and *b* are identical to the common ancestor, either because no change has occurred, or because two changes have occurred at the same site in the same branch (e.g. $A \rightarrow G$ and $G \leftarrow A$).

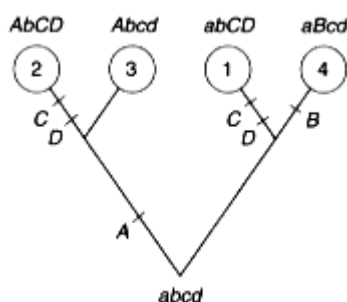


Figure 15.2

A tree that produces the genotypes listed in Figure 15.1, assuming that the change from *a* to *A*, or vice versa, was unique.

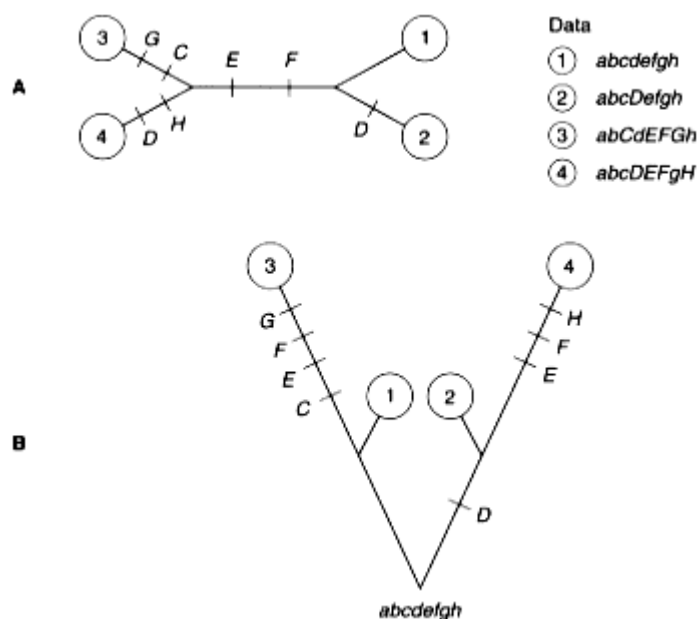


Figure 15.3

A tree with differing branch lengths. **A**, data, and the maximum parsimony tree; **B**, what actually happened. Because the branches leading to taxa 3 and 4 are longer (that is, more changes happened), and because some changes occurred independently in both branches, the maximum parsimony tree method leads to the wrong conclusion about the branching pattern.

2. Sites *c*, *d*, *g*, and *h* have changed in only one of the two branches. They are 'uninformative' in the sense described above, in that they do not influence the choice of the most parsimonious tree. However, they are not uninformative in a broader sense, because they tell us that taxa 3 and 4 are not closely related.

3. Sites *e* and *f* have changed in both branches. They are 'informative' but misleading, because they lead to the choice of the wrong tree.

Thus maximum parsimony may lead us to choose the wrong tree, either because not all changes are equally likely (e.g. synonymous versus non-synonymous changes), or because some branches are much longer than others. This has led to the idea that we should seek **maximum likelihood** tree. That is, we should first formulate some rules about how evolution has operated 根 or example, that transversions are rarer than transitions. We then seek that tree which, if it is true, maximizes the probability of getting the observed data.

Both the parsimony and maximum likelihood methods have an 'optimization criterion'. The likelihood method has the additional advantage that it forces us to think about what assumptions about evolution are appropriate to the data. Both methods, however, have the drawback that they are expensive in computer time. To look at all possible trees, and choose the optimal one (so-called exhaustive search) is ruled out for more than about 15 taxa. This has led to the development of a number of algorithms, or search rules, which seek, in the vast number of possible trees, for a smaller number of candidate trees, which can be tested against the optimization criterion. Alternatively, all attempt at strict optimization can be abandoned, and an algorithm can be adopted because it seems sensible, and is known to give satisfactory results on small data sets, to which optimization criteria can also be applied. One such method is the 'neighbour-joining' algorithm. This calculates a 'genetic distance', allowing for the likelihood of different changes, between all pairs of strains. It then selects two nearest neighbours, and replaces them by a single genotype that is a plausible ancestor of the two neighbours. By repeating this process until only two genotypes remain, a tree is constructed.

We must now return to the problem of how a tree is to be rooted. For molecular data, the usual method is to include an 'outgroup' in the analysis (Fig. 15.4). An outgroup is a related taxon known (or believed on good grounds) to have a common ancestor more distantly in the past than the taxa being classified. Thus, returning to the imaginary data shown in Fig. 15.1, suppose that, in addition to taxa 1-4, we have a fifth taxon, 5, differing from all of them at four further sites *e-h*. Because of these four differences, it is reasonable to assume that taxa 1-4 have a common ancestor more recently in the past than any of them do with taxon 5: that is, taxon 5 is an outgroup. For the sites varying within the taxa 1-4, taxon 5 has the genotype *abcd*. It is therefore parsimonious to assume that the common ancestor of taxa 1-4 also had this genotype, and to root the tree as in Fig. 15.1A. Of course, the root of the whole tree, including taxon 5, remains in doubt. Note that taxon 5 could have been identified as an outgroup by morphological rather than molecular data: for example, taxa 1-4 could be human, chimp, gorilla, and gibbon, and taxon 5 a baboon or macaque.

The Reliability of Trees

How reliable is an inferred tree? There are two kinds of error, systematic and random. A systematic error is one that would remain even if we had more data of

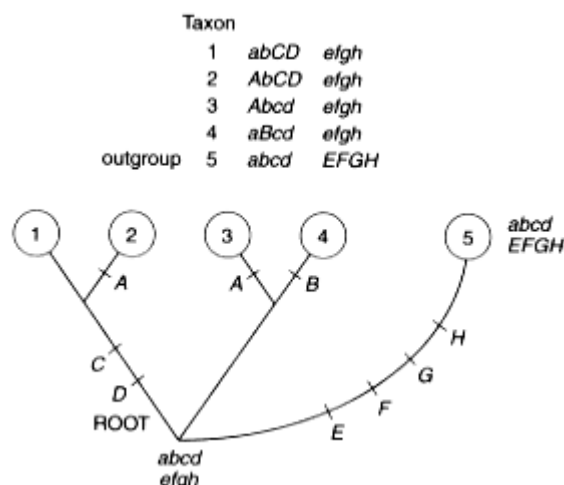


Figure 15.4
Rooting a tree.

the same kind. It arises because some inappropriate assumption has been made, explicitly or implicitly. Some possible sources of systematic error are:

1. There has been recombination or horizontal gene transfer *any* tree would be inappropriate.
2. Changes at different sites are not independent. This is particularly likely to be the case when using RNA molecules with secondary structure, because changes at one site in a hairpin loop will alter the likelihood of a change at the complementary site.
3. Excessive homoplasy. This is likely to be important when using maximum parsimony methods if branch lengths are very unequal, as illustrated in Fig. 15.3. Unequal branch lengths imply unequal rates of evolution, which in turn suggests that there has been directional selection: the difficulty is less likely to arise if changes have been neutral or nearly so.

Random errors arise because too few data are available. Suppose that a tree shows that man and chimp are more closely related than either is to the gorilla. How can one decide on the statistical significance of this conclusion? More generally, how can one decide on the significance of a particular branch point in a tree? The usual way of answering this question is the **bootstrap** method. Suppose, for example, there are 30 variable sites in the full data set. One constructs a new 'pseudo-sample' by sampling with replacement: that is, one chooses a site at random, adds it to the sample, and then replaces it in the full data set. One repeats this procedure until one has a new set of 30 sites. This new data set will contain some of the original sites once, some more than once, and some not at all. Having drawn such a sample, one constructs a tree, and notes whether it has the branch

point one is interested in. The 'bootstrap value' is the proportion of such trees that have the relevant branch point.

A high bootstrap value gives some confidence in the reality of the branch point. However, two warnings are necessary. First, the method gives no guarantee against systematic errors. Secondly, one should specify which branch point one is interested in before carrying out the test, because *some* branch point is likely to be preserved simply by chance.

What use are Phylogenetic Trees?

Three factors have made the construction of phylogenetic trees more reliable: the availability of molecular data, the use of computers to handle large bodies of data, and the development of a new theory of classification. But what can such trees tell us about the process of evolution? Some examples will help to make this clear:

1. The role of recombination. This is of particular importance in the study of the evolution of prokaryotes, because sexual fusion is never a requirement for reproduction, as it usually is in eukaryotes. The importance of recombination, therefore, is by no means obvious, and has in fact turned out to be very different in different kinds of bacteria. Phylogenetic analysis is useful in several ways. Suppose, for example, that sequences are available of several genes from a set of related strains. A phylogenetic tree can be constructed for each gene separately. If the trees are similar for the different genes, this suggests that recombination between strains has been infrequent. It is also possible to test for recombination by examining the sequences of the same gene from a set of strains. For example, the *recA* gene was sequenced from 15 strains of *Neisseria*, the genus to which the causative agents of gonorrhoea and bacterial meningitis belong. Thirty-nine synonymous third sites were polymorphic in the data set, each with two alleles. The most parsimonious tree had 18 homoplasies, compared to an expected number of 3.5 if there had been no horizontal transfer within the gene. There is therefore good reason to think that intragenic recombination has occurred. In such a case, the point of constructing a tree is not that it gives a correct picture of what has happened, but that it shows that no tree, representing a pure branching process, can be a correct picture.
2. The mechanism of speciation. Figure 8.9 (p. 153) superimposes a phylogeny of the deer mouse, *Peromyscus*, on a map of North America. This can tell us about the way in which geographical variation has arisen within a species, and can sometimes reveal incipient speciation.
3. The comparative method. Often, we try to identify the selective forces responsible for the evolution of some characteristic by comparing species. For example, in many primates, males are larger than females. Is this the result of competition between males for females, or an adaptation for defence against

predators? The first explanation is almost certainly correct, because the degree of sexual dimorphism in size and weapons is small in monogamous species, and large in polygynous ones. However, the conclusion depends on numbers; if only two primates were known, one monogamous and with no size dimorphism, and the other polygynous with a big size dimorphism, it would be unwise to draw any conclusion. What we need to know is the number of occasions on which the association between a particular trait (e.g. size dimorphism), and a particular ecological situation (e.g. polygyny) has evolved independently. There are problems in deciding on the relevant numbers. For example, there are several species of gibbon, all monogamous, and all with little difference in size between male and female. They should, therefore, count as only a single example of the coevolution of the two traits. Clearly, such decisions depend on an accurate phylogeny.

That the difficulty can be a real one is illustrated by Fig. 12.4, p. 232, showing that the chiasma frequency is higher in domestic animals, and supporting the theory that directional selection causes an increase in recombination rate. A weakness of the argument is that, of the seven domestic species, two were carnivores, four artiodactyls, and one a perissodactyl, whereas the wild species were marsupials, rodents, or primates. It would be good to have data on a zebra, an antelope, and a hunting dog. Clearly, inferences from the comparative method depend on reliable phylogenetic information. The matter is discussed further by Harvey and Pagel (1991).

Further Reading

Harvey, P.H. and Pagel, M.D. (1991). *The comparative method in evolutionary biology*. Oxford University Press.

Hillis, D.M., Moritz, C., and Mable, B.K. (ed.) (1996). *Molecular systematics* (2nd edn). Sinauer, Sunderland. (Particularly the chapter on phylogenetic inference.)

Ridley, M. (1986). *Evolution and classification*. Longman, New York.

Answers to Problems

Chapter 1

1. $26^{19} = 7.66 \times 10^6$.
2. (a) $P(\text{all 19 letters wrong}) = (25/26)^{19} = 0.475$, so $P(\text{at least one letter correct}) = 0.525$. (b) $19 \times (25/26)^{18} \times 1/26 = 0.361$.
3. $P(\text{a particular incorrect letter mutates to a correct one}) = 1/100 \times 1/25 = 0.0004$. Hence $P(\text{in all 10 copies, all letters remain incorrect}) = (1 - 0.0004)^{10} = 0.9268$. Hence $P(\text{at least one letter correct}) = 0.0732$.
4. $1/0.0732 = 14$ generations approximately.
5. For a particular copy, $P(19 \text{ are correct}) = P(18 \text{ remain correct}) \times P(19\text{th mutates to correct value}) = (99/100)^{18} \times 0.0004 = 0.000334$. Hence $P(\text{one copy out of 10 has all 19 correct}) = 0.00334$.
6. Perhaps it is the fact that the program has a representation of the optimum message, and determines the 'fitness' of actual messages by comparing them to the optimum. No analogous process occurs during natural selection.

Chapter 2

1. $\delta x = x(r\delta t - cx\delta t)$, or $dx/dt = rx(1 - cx/r)$. The carrying capacity $K = r/c$. Note that the carrying capacity is reached when births = deaths: that is, when $r = cx$. Referring to Box 2.1, this can be regarded as microscopic justification of the logistic equation.
2. (a) Probability of accurate replication $Q = 0.999^{100} = 0.9048$. (b) $(RQ - r)x_1 = R(1 - Q)x_0$, or $(Q - r/R)x_1 = (1 - Q)x_0$. If $x_0 + x_1 = 1$, $Q = 0.9048$ and $r/R = 1/1.2$, then $x = 0.429$. (c) If u is the per base error rate, then $Q > r/R$ requires $(1 - u)^{100} > 1/1.2$, or $u < 0.00182$.
3. Given randomness, the following should be true:
 - (i) The expected number of generations in which the copy number increases equals the number in which it decreases. For n generations, the variance of the number is $npq = n/4$.
 - (ii) Following an increase (or decrease), the copy number is equally likely to increase or decrease.
 - (iii) The change in copy number in a generation should be normally distributed, with mean 0 and variance $50p(1 - p)$, where p is the frequency in the preceding generation.

4. The expected number in the next generation is normally distributed, with mean 700 and standard deviation $\sqrt{(1000 \cdot 0.7 \cdot 0.3)} = 14.5$. To exceed 730 is to be just over twice the SD above the mean, which will happen with a probability of approximately 1/50. (A Poisson family-size distribution has been assumed.)
5. There are 1000 gene copies. The expected time to fixation of one copy by drift is $2N = 2000$ generations.
6. Whichever type is initially more abundant will eliminate the other.

Chapter 3

1. Not consistent (χ^2 with 1 df = 7.9, $P < 0.01$). There is an excess of homozygotes. This could be caused by selection, but a more likely explanation is that the mice do not come from a single random-mating population. If so, other loci should show a similar departure from the Hardy-Weinberg ratio.
2. Ginger, 0.0025; tortoiseshell, 0.095. Assumptions: equal gene frequencies in males and females, random mating.
3. No. The expected numbers are 243.9 of each homozygote, and 512.2 heterozygotes. Actual numbers would, of course, differ from these values by chance. But the difference between the expected numbers, and those expected from the Hardy-Weinberg ratio (250:500:250) is far too small to be detected.
4. Let the frequency of one allele, A , be p in males and P in females. Then the frequency of AA among the progeny is pP . The frequency of A among the progeny is $(p + P)/2$, and so the frequency of AA expected from the Hardy-Weinberg ratio is $(p + P)^2/4$. Hence (actual frequency of AA) - (frequency of AA expected from the Hardy-Weinberg ratio) = $pP - (p + P)^2/4 = -(p - P)^2/4$. Hence, if $p \neq P$, there is a deficiency of homozygotes, and therefore an excess of heterozygotes.
5. (a) 0.39 A:0.24 B:0.12 AB:0.25 O. (b) The mother must be BO ; the father is AA with probability $0.09/0.39 = 0.231$, and AO with probability 0.769. Hence the sib is A with probability $(0.231 \cdot 1/2) + (0.769 \cdot 1/2 \cdot 1/2) = 0.308$.
6. Type A , $R_0 = 2$, $r = 0.1386$; type B , $R_0 = 3$, $r = 0.1569$. (Remember that only one half of the seeds are female.) Since both the values of r are positive, the population is increasing, regardless of the frequencies of the two types. Type B is fitter.
7. (a) Since p and s are both small, we can use $p_n = p_0 e^{shn}$, where $s = 0.01$, $h = 1$, $p_0 = 1/100\,000$, and $p_n = 1/100$. Hence $e^{0.01n} = 1000$, or $n = 691$ generations. (b) Integrating $\dot{p}/dt = sp^2$ gives $st = \text{Const.} - 1/p$. When $t = 0$, $p = 1/100\,000$, and hence $st = 100\,000 - 1/p$, or $t \approx 10^7$ generations. The result is misleading for the following reason. It assumes that, if $p = 1/100\,000$, the frequency of homozygotes is 10^{-10} . Now a very large population is unlikely to mate randomly, because most individuals will mate with others that were born reasonably close

to them (this may not be true for marine species with planktonic larvae), and therefore their mates will be more closely related to them than a random member of the species. Therefore, the frequency of homozygotes will be higher than predicted from the Hardy-Weinberg ratio. However, it is true that the initial increase of a rare favourable recessive will be very slow, unless inbreeding is common. The problem of structured populations is discussed further in Chapters 8 and 9.

Chapter 4

1. At the first locus, the heterozygosity is $2[(0.12 \times 0.7) + (0.7 \times 0.18) + (0.18 \times 0.12)] = 0.4632$; at the second locus, it is $2 \times 0.37 \times 0.63 = 0.4662$. It is zero at the other two loci. Hence average heterozygosity = 0.232.

2. (a) $26 \times 25/2 = 325$. (b) If we look at the 26 lethals in turn, the probability that the second is different from the first is 0.999; if so, the probability that the third is different from the first two is 0.998; hence the probability that all 26 are different is $0.999 \times 0.998 \times \dots \times 0.975 = 0.721$. (c) Probability that exactly one pair of alleles found = (probability that a particular pair are alleles) $\times$ (number of pairs) $\times$ (probability that remaining 24 are all different) = $(1/1000) \times (26 \times 25/2) \times 0.758 = 0.246$. (d) Suppose that an essential protein is coded for by two (or more) linked loci. If the frequency of non-functional recessives at each locus is low, there is almost no chance of finding individuals homozygous for non-functional alleles at both loci. Hence our estimate of the number of essential genes will not include proteins coded for by a gene family. But will the frequency of non-functional recessives be low? It will, provided that there is some difference between the loci, so that individuals lacking one or other of the proteins, although viable, are less fit than individuals with both loci functional. If the loci code for identical proteins, one of them is likely to be lost by the accumulation of mutations.

3. $u = psh$. $p = 10/2 \times 94 \times 0.75$. Fitness of heterozygotes = $27/108 \times 457/582 = 0.20$. Hence $hs = 0.80$, and $u = 4.3 \times 10^5$.

4. (a) $p = \sqrt[4]{(u/s)}$, or $u = p^2 s$, where $p^2 = 1/10 \ 000$ and $s = 1$. Hence $u = 1/10 \ 000$. (b) if the relative fitnesses are $1:1-\delta$, then from Equation 4.5 $p = \delta/(1 + \delta)$. Since $p = 1/100$, $\delta = 1/99$.

6. $p(A) = 0.9091$; $W = 0.9545$, so load = 0.0455.

7. Fitness of homozygote = 0.889; $W = 0.90$, so load = 0.1.

8. At equilibrium, $0.75 + P = 1.5 - P$, or $P = 0.375$. If p is the frequency of A , then $(1 - p)^2 = 0.375$, or $p = 0.388$. The equilibrium is stable, because dark is fitter than pale when dark is rare, and pale is fitter than dark when pale is rare. The genetic load at equilibrium is zero, because all three genotypes have the same fitness.

9. Relative to M flies, the viability of $+^1/+^2$ is $467/2 \times 201 = 1.1617$, and of $+^1/+^1$ is $376/2 \times 197 = 0.9543$. Hence the relative viability of $+^1/+^1$ is 0.821.

10. If there are no cell deaths, one cell gives rise to 10^8 cells by $10^8 - 1$ cell divisions. If the back mutation rate = u , the probability of no back mutations in a tube is $(1 - u)^{10^8} = \exp(-10^8 u) = 0.72$, or $u = 3.28 \times 10^{-9}$.

Chapter 5

1. -0.02.

2. $-0.02 - 0.9 = -0.0131$.

3. Yes. Writing the frequencies of the four phenotypes as $p(AB)$, $p(Ab)$, $p(aB)$, and $p(ab)$, we expect $p(AB)p(ab) = p(Ab)p(aB)$ if there is linkage equilibrium.

4. (a) Zero. (b) Negative. Let the initial frequencies of the haplotypes be $p(AB)$, $p(Ab)$, $p(aB)$, and $p(ab)$, where $p(AB)p(ab) = p(Ab)p(aB)$. In case (a) D after one generation is $p(AB)p(ab) - [(1+s)(1+t) - (1+s)(1+t)] = 0$. In case (b) $D = p(AB)p(ab) - [1+s+t - (1+s)(1+t)] = -st p(AB)p(ab)$. It is an important result, due to Felsenstein, that, if a population starts in linkage equilibrium, and fitnesses are multiplicative (case a), it remains in linkage equilibrium. But if the population is exposed to directional selection, and fitnesses increase less steeply than in the multiplicative case, then negative linkage disequilibrium is produced.

Chapter 6.

1. (a) 6.5 $\times 10^5$; (b) 2.44.

2. Heritability = $2.44/6$, so response = $1 - h^2 = 0.407$.

3. Sex is genetically determined, but there is no correlation between the sex of a child and a parent. So the broad-sense heritability = 1, and the narrow-sense heritability = 0.

4. If the half-sibs were raised apart, this implies a heritability of $4 - 0.3 = 1.2$, which is not possible. But if they are raised together, or in similar environments, a value of 0.3 is quite plausible.

5. 1.4.

6. $h^2 = 0.4$, and $S = 3$, so expected value of offspring is $18 + 0.4 \times 3 = 19.2$ bristles.

7. Total variance = 9, and $h^2 = 0.3$, so $V_A = 2.7$. Total difference $D = 16$. From Box 6.7, if initial gene frequencies are 0.5, number of loci $= D^2/\delta V_A \cong 12$ loci.

8. 0.63ab:0.37AB. Note that this implies instability of the equilibrium shown in Table 6.5, because the frequencies have moved away from the equilibrium point. Usually, epistatic fitnesses give rise to alternative stable monomorphic states, with an unstable equilibrium between them. Stable polymorphic states, such as heterostyly and mimicry, are maintained by frequency-dependent fitnesses.

9. If there are n loci, and the effect per allele substitution is d , then the initial

variance is $n \cdot 2 \cdot 0.01 \cdot 0.99 = 0.0198d^2$. The maximum variance is reached when half the loci have $p = q = 0.5$; by this time the remaining loci will contribute little. Hence the maximum variance $n(2) \times 2 \cdot 0.5 \cdot d^2 = 0.25nd^2$. That is, the genetic variance is increased by a factor of 12.6. In practice, the increase would be less than this, because different loci would change at different rates, but it should still be detectable.

Chapter 7

1. (a) All R , or all S . (b) The unstable equilibrium, with R and S equally fit, occurs when $p(R) = 0.25$, $p(S) = 0.75$. If R is initially commoner than 0.25, it will increase. When $p(A) = 0.4$, $p(a) = 0.6$, and hence, with random mating, $p(R) = 0.36$. Hence the population will evolve to $p(R) = 1$, $p(a) = 1$.
2. The ESS is $p(R) = 0.25$, $p(S) = 0.75$. Therefore $p^2(a) = 0.75$, or $p(a) = 0.866$.
3. There are two ESSs: all C , or $M = (0.5A:0.5B)$. Note that $E(M, M) = 3.5$, and $R(C, M) = 2$, so C cannot invade M .
4. Let FA , FL , and FS be the strategies 'always escalate', 'escalate if larger, withdraw if smaller', and 'escalate if smaller, withdraw if larger', respectively. In a contest, an individual has a 50 per cent chance of being larger. (a) $E(FA, FA) = R/2 - C/2$; $E(FL, FA) = 1/2[PR - (1 - P)C]$. Hence FA is stable against invasion by FL provided that $R - C > PR - (1 - P)C$, or $(1 - P)R > PC$. This is possible provided that $P \neq 1$, and the reward is large relative to the cost. (b) $E(FS, FS) = R/2$; $E(FA, FS) = 1/2[RP - (1 - P)C] + R/2$. Hence FS is stable against invasion by FA provided that $RP - (1 - P)C < 0$, or $RP < (1 - P)C$. Hence the paradoxical strategy, 'escalate if smaller', can be an ESS, provided that $P \neq 1$, and costs are large relative to rewards. But I doubt whether such a paradoxical strategy has often evolved. Whenever the payoffs are such as to make it possible, the 'common-sense' strategy 'escalate if larger' is also an ESS. (c) The 'payoff' for death means the change in fitness caused by death. Hence C is the expected reproductive success of an animal that withdraws from the contest without fighting.
5. (a) The payoff matrix is:

	watch	don't watch
watch	4	4
don't watch	5	2.5

Hence the evolutionarily stable state is 3/5 watch: 2/5 don't watch. (b) Watchers always have a fitness of 4. Non-watchers have a fitness of 5 when rare, and 2.5 when common, so there must be a mixed ESS. Let p be the proportion of watchers at the ESS. Then the chance that a non-watcher will flock with four other non-watchers is $P = (1 - p)^4$. Now at the ESS watchers and non-watchers are equally fit, so $4 = 5(1 - P) + 2.5P$, or $P = 2/5$. Hence $p = 0.205$.

Chapter 8

1. There is a chance of $1/16$ that an F_2 individual will be homozygous for a particular lethal present in a grandparent, and $15/16$ that it is not. Since there are 12 lethals expected in the grandparents (4 genomes $\times$ 3), the probability that an F_2 animal survives is $(15/16)^3 = 0.461$.

2. (a) Yes. The chance that a particular locus is still heterozygous is $1/2 = 0.5$. Hence the chance that it is homozygous at all loci is $(15/16)^6 = 0.679$, or greater than one-half. (b) The chance that an F_4 plant is heterozygous at all six loci is $(1/16)^6 = 6 \times 10^{-8}$. The chance that it is heterozygous at five loci is $6 \times (1/16)^5 \times 15/16 = 5.36 \times 10^{-6}$. So the chance of a plant being as (or more) heterozygous as the one examined is 5.42×10^{-6} . This is the kind of thing that doesn't happen to me, so I would conclude that the simple theory does not apply, probably because there is selection favouring heterozygotes.

3. $N_e = 21.4$.

4. (a) $2s = 1/50$. (b) From Equation 3.6, the expected number of copies of a gene after 60 generations is $n_{60} = \exp(0.01 \times 60) = 1.822$. However, if n is the number of copies, given that it survives, and P is the probability that it does survive, then $n_{60} = nP + 0(1 - P)$ or $n = n_{60}/P = 1.822 \times 50 = 91$. (Note that this assumes that a gene that has survived for 60 generations is sure to be established.)

5. Taking a human generation as 20 years, the evolution rate is $0.15 \times 20 \times 10^6 = 3 \times 10^6$ substitutions per gene per generation. On the neutral theory, this should equal u , the neutral mutation rate. The expected proportion of the population homozygous is $1/(1 + 4N_e u)$, or approximately 1 in 10^6 . The observed proportion is greater than 0.99. However, this discrepancy does not disprove the neutral theory, because the expected homozygosity is an equilibrium value, approached after N_e generations. The observations are consistent with the neutral theory if the human population has recently passed through a bottleneck of numbers.

Chapter 9

1. No. The genotype of this 'species' consists of those genes that happened to be present in the original hybrid.

2. Yes. A gene causing sib altruism in armadillos would spread if $b > c$, compared to the requirement $b > 2c$ in typical litter-producing mammals. So far as I know, altruism towards sibs has not been observed in armadillos. The explanation may be that the biology of armadillos is such that no mutations favouring sib altruism have occurred. This is to explain the absence of altruism by appealing to 'developmental constraints'. This may be correct, but unless one can point to specific features of armadillo biology that make sib altruism impossible, it is an unhelpful explanation.

3. (a) More than 3, that is, 4; otherwise the subordinate would do better to leave, and one would not observe pairs. (b) None. Calculate the number N of genes transmitted that are IBD to the subordinate if she leaves, and if she stays. If she leaves, $N = 3$ (her own offspring) + $3/2$ (her dominant sib's offspring) = 4.5. If she stays, $N = 0$ (her own) + $10/2$ (her sib's) = 5. So it pays the subordinate to stay, even if she raises no eggs of her own (but it wouldn't if there was any appreciable doubt that the dominant bird was her full sib).

4. (a) Yes. The dominant would raise 5 offspring instead of $7/2 = 3.5$. (b) No. The dominant has an inclusive fitness of 5 if he drives his brother out, and $7/2 + 1/2 \cdot 7/2 = 5.25$ if he does not. (c) Yes. The dominant has an inclusive fitness of $7/2 + 1.4 \cdot 7/2 = 4.375$ if he allows his half-brother to stay, which is less than 5.

5. The female would not join her brother. If N is the number of genes IBD to her own that are transmitted to the next generation, then if she joins $N = 1/2$ (her own genes) + $1/2 \cdot 1/2$ (her brother's genes) = $3/4$, whereas $N = 1$ if she does not join. The male would accept his sister if she did join: from his point of view, $N = 1/2 + 1/2 \cdot 1/2 = 3/4$ if she joins, and $1/2$ if she does not.

6. $9/16$ of females comes from joint nests. If they do, their probability of being full sibs is $0.8 \cdot 0.2^2 = 0.032$. If they are not full sisters, their coefficient of relatedness is $1/2$. Hence $r = 7/16 \cdot 0.75 + 9/16(0.68 \cdot 0.75 + 0.32 \cdot r/2)$, or $r = 0.676$.

Chapter 13

Part A

1. Female bias (sons cost more).
2. No bias.
3. Male bias (male zygotes are cheaper).
4. No bias.
5. No bias (eggs cost just as much to lay even if they die).
6. Female bias (the colony is the unit of evolution, and is more likely to split into two if it contains more females).
7. Females produce two sons, or two daughters, with equal frequency: there are no mixed litters.

Part B

1. A single seed can found a population.
2. By suppressing male function, a cytoplasmic gene increases seed production, and hence its own transmission to the next generation.
3. In haplo-diploids, recessive genes are exposed to selection in males, on average once in three generations. Hence the load of deleterious recessives in an outbred population will be lower in haplo-diploids, and therefore inbreeding will be less harmful.

4. In a large plant, selfing between different flowers is more likely than in a small one.
5. Most animals have separate sexes. Allopolyploidy would mess up most sex-determining mechanisms. An alternative explanation is that tetraploid cells have twice the volume of diploid ones, and a doubling of cell volume would be more damaging in an animal than a plant.

Chapter 14

1. Number of generations = 2.67×10^5 . If volume increased by a fraction x per generation, then $(1 + x)^{2.67 \times 10^5} = 3$, or $x = 0.4115 \times 10^{-5}$. Hence $S = 0.823 \times 10^5$ mean volume. Hence $I = 0.823 \times 10^4$.
2. A simple method is as follows. Did the ratio (number of artiodactyls/number of perissodactyls) increase in all nine time-intervals? The probability of a change in the same direction (up or down) in nine successive intervals, on the null hypothesis, is $1/256$.
3. It could be that taxonomists have placed any fossil they believe to be ancestral to *Oxodonta* in that genus, and similarly for the other two genera. If so, and if evolutionary divergence has proceeded at a uniform rate, the morphological differences between early representatives of the three genera should be less than between recent ones. To test this, you would have to devise a measure of morphological difference.

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